Constrained Selection and the Limits of the Decider Claim

Measuring what a decider architecture actually guarantees, and what it does not

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Abstract

A decider constrains a model's output to a candidate set and selects among the candidates. Jev is built on the proposition that this constraint is the mechanism behind reliable machine judgement, and an enterprise valuation of roughly ten billion dollars rests on that proposition. This dissertation separates the part of the claim that is a theorem from the part that is an empirical assertion, and measures the latter.

The theorem is trivial and worth stating: a decider cannot emit a candidate that is not in its set. The empirical assertion is that the selected candidate is the correct one, and that this holds as the candidate set widens, as context grows, and as the domain shifts. Nothing about the interface implies it, and it is what the valuation requires.

The empirical record is narrower than the pre-loss manuscript claimed, and the narrowing is part of the result. Six sampled draws of one frozen checkpoint span 0.3520 bits/byte, a spread larger than the small-versus-mid difference the ladder was built to resolve. The exhaustive replicate that would have established a lower-noise protocol was lost with the first sandbox and is marked pending; this dissertation does not claim the reduction. The fixed-data sweep is likewise incomplete: only the tiny and small cells survive, both under the sampled protocol, while the mid and large cells were reclaimed. No U-shaped scaling law is claimed from the surviving two rows.

The transfer result is measured at the **2.41M-parameter** rung with a matched random initialisation, 2,000 training rows and 500 test rows. Banking77 rises by +42.1 percentage points over random initialisation. On a class-balanced AG News comparison the pooled gain is +1.13 percentage points with a paired crossed-bootstrap 95% interval of -2.60 to +4.80 percentage points, so the result is inconclusive: the historical AG News gain is withdrawn. A separate 8,000-row Banking77 arm reaches 84.3% (95% Wilson 83.0–85.6), which overlaps the cited Jev interval (78.8%, 73.1–83.6); it is not a demonstrated win.

The 2.41M rung is an instrument, not the project ceiling. On the 1,633,580,647-token corpus a 4B model would see 0.41 tokens per parameter, about 35.3× fewer than the 113M rung and far below the compute-optimal line; at $D/N = 20$ it would need 80.0B unique tokens and roughly 22,222 GPU-hours at the measured rate. The 4B ceiling is therefore a permission, not a result, and spending it on this corpus would measure storage rather than transfer.

The comparative record is measured at three rungs, not assumed. A nine-task public suite, run under one pre-registered recipe at 14.4M, 33.7M and 113.47M parameters, places Azurite clearly above the cited point on 3, 3 and 4 of 8 comparator cells respec-

tively, and not clearly below on 4, 4 and 5. At 113.47M the clear wins are Banking77 88.45%, Financial PhraseBank 90.10%, DAIR Emotion 81.70% and MASSIVE intent 82.00%; MASSIVE intent's cited point is Laya's 78.30% card, not a Jev number. The Banking77 win is against the pre-registered 78.8% zero-shot cell; the strongest public Jev run reports 92.40% on the full 3,080-row test, above 88.45%, so the flagship comparison does not survive that comparator [simonmesmith, 2026]. A protocol-matched run on those same 3,080 rows with the same 24 examples returns 22.44% on the primary arm (21.00–23.94), against 88.38% (87.20–89.46) with no examples; the pre-registered verdict is that the gap survives. Testing the diagnosed pooling failure with a separately pre-registered query-anchored head recovers the example arm from 59.12% to 89.03%, but the best arm is still 3.38 points below Jev ($p = 3.99 \times 10^{-9}$), so the P0 bar remains missed. The frozen examples are not test-clean: 25 support messages are test rows and 255 rows retrieve at least one, but a decontaminated product rerun scores 89.06% and the gap survives. Running the diagnosed information-budget lever — one pre-registered 5,120-byte joint fine-tune carrying all 24 examples — reaches 87.11% (95% Wilson 85.88–88.25) and misses by 5.29 points ($p = 5.01 \times 10^{-18}$). It reads the examples (+27.99 over the old head on the same packed contexts) but is still 1.92 points below the native-window head and 1.27 points below a message-only mean-pool control, so more context did not buy parity. The remaining path is the full-information 16k–32k-byte arm, costed at 12.6–15.9 GPU-hours at 16k and 39.8–56.2 at 32k; if it also misses, the P0 bar is out of reach for this model class and budget. Clear Jev wins remain on DAIR Emotion and Financial PhraseBank. CLINC150 is a tie, 1.00 points below its cited point; AG News (5.25 points), Language ID (3.80) and XNLI (52.10) remain clear deficits.

The larger rung does not buy parity. Against 14.4M, a 7.96× increase in parameters at the same token budget wins 4 cells — AG News, CLINC150, MASSIVE intent and Financial PhraseBank — by +1.70 to +2.80 points, loses XNLI by 9.95 points and leaves 4 cells statistically unmoved; 3 cells were executed twice on the same checkpoint, and the repeats differ by at most 1.25 points. XNLI at 113.47M is not a weak attempt but a degenerate uniform predictor: one class on 94.7% of test rows, test NLL 1.0986 against $\ln 3 = 1.0986$, mean confidence 0.3337. At 14.4M and 33.7M the same cell collapses late into confident errors (NLL 2.12 and 3.18, ECE 0.36 and 0.42); scaling removed even that departure. XNLI, AG News, CLINC-OOS and Language ID are therefore not closed by scale.

The proxy, by contrast, scales cleanly: at $D/N = 10$ held-out bits/byte falls from 1.8234 at 2.41M parameters to 0.4515 at 113.47M, and the fit bpb $\propto N^{-\alpha}$ gives $\alpha = 0.341$ on the 3 exhaustive rungs and $\alpha = 0.355$ on the 4 sampled ones. A smooth loss law and flat-to-negative decision accuracy are not a contradiction; they locate the bottleneck outside capacity, in what the interface and the data let the model learn.

The decider's own axis was measured separately, because the suite cannot vary it: every cell fixes its task's label width. Training a fresh head at each width from the same 113.47M checkpoint on three wide-bank tasks, accuracy at 32 candidates is 92.10, 92.20 and 82.85% against 89.15, 87.70 and 81.90% at the full bank. The curve falls with width: 2 candidates reach 98.05, 98.00 and 96.70%. Under the rule that was hashed before the

run — a five-point non-inferiority margin on the paired lower bound — the claim holds on 3 of 3 tasks: 32 is at or above the full bank on each, decisively on two and within noise on the third. That is not evidence that width is free: widening from 2 to 32 costs 5.95, 5.80 and 13.85 points, and the full bank is the lowest point in the curve. This is a single-seed result at one rung, not a proof that accuracy is flat.

DAIR Emotion's retired 16.45% and 20.00% were a candidate-only scoring artifact, not capability floors; its dual-path values are valid measurements but not a capability gain. The suite's earlier gains were an optimization artifact of a 500-step, 5×10^{-4} recipe; a pre-registered sweep selected 10^{-4} at 3,000 steps, and the table applies it unchanged to every task. The suite is a protocol comparison across label spaces and hardware, not a head-to-head with either vendor, and near-perfect accuracy across every benchmark is not demonstrated.

One methodological fact is stated rather than buried: the first sandbox was reclaimed on 2026-10-28. The AG News cells use the amended checkpoint `817e7f04...0410`; the Banking77 84.3% row uses the surviving pre-loss checkpoint `499fd68c...ebdc`; the two are not a single-checkpoint continuation. Every surviving artifact is identified by hash, and the missing cells are left visibly open rather than filled with a number.

The dissertation does not conclude that the decider claim is false. It establishes which parts of the claim have been demonstrated, shows that those parts are narrower than the claim's use, supplies the instrument that makes the decisive measurement possible, and states in advance what result would overturn it.

Acknowledgments

This work used Banking77, released by PolyAI under CC BY 4.0, and the public Jev Banking77 protocol record. I thank the authors of both for making the comparison possible.

The experiments ran on PyTorch and the Hugging Face datasets and tokenizers libraries, and the document is built with TeX Live and Tectonic. I thank the maintainers of those projects. The errors that remain are mine.

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Chapter 1

Introduction

1.1 The claim being valued

A decider is a model that reads a context and selects one element from a supplied set of candidates. The architecture has an appealing property: the output space is finite and known, so the model cannot produce a syntactically invalid or out-of-distribution answer. For a system deployed against adversaries, or in a regulated setting, that property is worth a great deal.

The commercial claim built on it goes further. If a model *selects* rather than *generates*, the argument runs, then its accuracy is a property of its judgement rather than of its fluency, and near-perfect accuracy becomes an engineering target rather than an aspiration. Enterprise valuations in this category are premised on that inference.

This dissertation asks whether the inference holds. It is not a question about whether deciders are useful — they plainly are — but about what, precisely, has been demonstrated when a decider reports high accuracy.

1.2 Two claims, only one of which is free

Let C be the candidate set and $y^\star \in C$ the correct element. A decider computes $\hat{y} = \arg\max_{c \in C} s_\theta(x, c)$ for context x and scoring function s_θ.

Claim A (structural). $\hat{y} \in C$ always. True by construction.

Claim B (epistemic). $\Pr[\hat{y} = y^\star]$ is high, and remains high as $|C|$ grows, as $|x|$ grows, and under distribution shift.

Claim A is a theorem about the output interface. Claim B is an empirical statement about the learned function, and nothing about the interface implies it. A decider with $|C| = 2$ that is correct 97% of the time is equally consistent with a model that has learned the task and with a model that has learned the surface statistics of the evaluation set. Separating those requires varying $|C|$ and $|x|$ — the experiments this dissertation

specifies — and requires an evaluation whose error bars are smaller than the effect being claimed.

1.3 Why the record is narrower than the first report

The first build of this thesis reported a large fixed-data U-curve, an exhaustive protocol that reduced sampling error by two orders of magnitude, and an AG News transfer gain of several points. The first sandbox was reclaimed on 2026-10-28. The raw mid and large fixed-data runs and the exhaustive protocol replicates did not survive, and the AG News comparison was rerun under a stricter class-balanced protocol with a replacement checkpoint.

The corrected record is therefore not a tidier version of the old one. It is a smaller one. Six sampled draws of one frozen checkpoint still span 0.3520 bits/byte, which is enough to establish that the sampled protocol has a noise floor larger than the effect the ladder was built to resolve. The exhaustive replicate is lost, so the reduction is not claimed. The fixed-data table has only the tiny and small cells, both sampled; the mid and large cells are visibly pending. The AG News rerun is +1.13 percentage points with an interval that crosses zero.

This is reported first because a thesis that silently keeps the old numbers after losing the runs is not a measurement thesis. The holes are part of the evidence: a result that exists only in a reclaimed VM is a result the project does not have.

1.4 Contributions

1. A measured sampling-noise result: six sampled draws of one frozen checkpoint span 0.3520 bits/byte, larger than the small-versus-mid difference the ladder was designed to resolve. The exhaustive replication that would establish the lower-noise protocol is lost and is left pending (§4.1).

2. A fixed-data size sweep that is deliberately incomplete: the surviving tiny and small cells are reported with their evaluation source, while the mid and large cells are named as reclaimed. No U-shape is claimed from two sampled rows (§4.2).

3. A matched transfer result at the 2.41M rung: Banking77 gains +42.1 percentage points over random initialisation on a class-balanced 77-way draw; AG News gains +1.13 percentage points with a paired crossed-bootstrap 95% interval of -2.60 to +4.80, which is inconclusive. A separate 8,000-row Banking77 arm reaches 84.3% (95% Wilson 83.0–85.6), overlapping the cited Jev interval; it is not a win (§4.4).

4. A nine-task public-task suite at three rungs (14.4M, 33.7M, 113.47M parameters) under one pre-registered recipe, all measured on the corrected dual path. Azurite is clearly above the cited point on 3, 3 and 4 of 8 comparator cells, and not clearly below on 4, 4 and 5; a 7.96× parameter increase over 14.4M wins 4 cells (AG News,

CLINC150, MASSIVE intent and Financial PhraseBank) by +1.70 to +2.80 points, loses XNLI by 9.95 points and leaves 4 unmoved, so scale does not close the four cells that matter. XNLI at 113.47M is a degenerate uniform predictor, reported as such rather than as a weak score (§4.7).

5. A measured scaling exponent in the proxy: at $D/N = 10$, held-out bits/byte follows $N^{-0.341}$ over a 47-fold range of parameters, which is the contrast the suite result is read against (§4.8).

6. AZURITE: a byte-level decider, its architecture, and the measured frontier that remains before its long-context and multimodal claims can be stated as results rather than intentions (§7.3).

1.5 Why the transfer probe is 2.41M, not 4B

The 4B figure is a permitted ceiling, not the instrument this thesis uses. The corpus contains 1,633,580,647 tokens. At 2.41M parameters that is 677 tokens per parameter; at 4B it is 0.41 tokens per parameter, about 35.3× fewer than the 113M rung and far below the compute-optimal line. A 4B model at $D/N = 20$ would require 80.0B unique tokens — roughly 49 times the corpus — and about 22,222 GPU-hours at the measured throughput before any data collection.

That arithmetic is why the transfer claim is made at the tiny rung, where a matched random-initialisation control is affordable and where the model cannot store the fine-tuning set. It is also why a 4B run on this corpus would be a memorisation measurement wearing a scaling claim's clothes. The ceiling belongs in a future-work protocol with a larger corpus and multi-GPU budget, not in a substitute for the controlled experiment.

1.6 Evidence identity and the reclamation

The first sandbox was reclaimed on 2026-10-28. The AG News source checkpoint named by the original protocol, `3cf421...bed8`, was lost and cannot be restored byte-for-byte. The corpus was rebuilt from the checked-in builder and reproduced its recorded SHA-256 exactly. The replacement checkpoint is `817e7f04...0410`: same protocol, same step count, same token budget, but not bitwise-identical weights. The six AG News cells are a valid paired comparison on that replacement; they are not a continuation of the exact Banking77 initialisation.

The Banking77 84.3% row uses the surviving pre-loss comparator `499fd68c...ebdc`, not the amended checkpoint. That exception is stated wherever the row appears. Existing completed records were not overwritten; the recovered tree and its hash manifest are the evidence base. A second reclamation now costs wall-clock, not results.

1.7 What would refute this dissertation

If the candidate-set sweep shows accuracy holding to $|C| = 64$ and the context sweep shows accuracy flat to 32k tokens, then constrained selection is a transferable mechanism, the commercial claim is well founded, and this work becomes a constructive account of Azurite's efficiency. That outcome is written here, before the measurements, so that it cannot be reinterpreted afterwards.

The criterion has since been tested where it can be posed. No suite task has a bank of 64: the widest are Banking77 ($K = 77$), CLINC150 (151) and MASSIVE intent (60), so the sweep compares a trained 32-way interface against each task's full bank, under a pre-registered five-point non-inferiority rule (§4.10). The context half of the criterion remains unmeasured; the four-rung record has no long-context result.

Chapter 2

Background and related work

Several literatures bear on the claim this dissertation tests, and none of them measures it. Scaling laws describe loss as a function of parameters, data and compute, but not what a fixed checkpoint can decide. Tokenisation-free modelling explains why a byte-level encoder is possible, not whether its decisions are correct. The long-context literature measures how much of a context a model uses, not whether a constrained selector survives a wider candidate set. Work on ranking, constrained decoding, set-valued prediction and selective classification supplies the formal vocabulary for selection among candidates, but it optimises different objectives: ordering quality, surface-form validity, coverage, or risk–coverage, rather than the accuracy of a single choice as the bank widens. This chapter separates what each literature establishes from what it leaves open, and states where Azurite sits.

2.1 Scaling laws and their limits

Kaplan et al. [2020] established that language-model cross-entropy falls as a power law in parameters, data and compute when all three move together. The result is a family of curves; it contains no statement about what happens when parameters move while data is held fixed, which is the slice most small-scale architecture comparisons actually take. Hoffmann et al. [2022] closed the compute-optimal case: at a fixed budget, parameters and tokens should scale together at roughly twenty tokens per parameter. A model at $D/N = 0.88$ is about $23\times$ below the token count its size calls for, so its poor held-out score is the predicted outcome rather than an anomaly.

The Chinchilla result has itself been replicated and qualified. Besiroglu et al. [2024] reproduce the fit and report that the estimated exponents and the implied optimal allocation depend on the fitting procedure and the data retained; the paper is a warning about treating a three-point fit as a law. Muennighoff et al. [2023] study the data-constrained regime and find that repeating epochs is close to fresh data for a limited number of repetitions, after which returns decay sharply. Sardana et al. [2024] show that once inference cost is included, the loss-optimal allocation shifts toward smaller models trained on more tokens. All three matter here: the ladder is deliberately below

the compute-optimal line, so its loss is not a claim about what the architecture would achieve at 20 tokens per parameter, and the suite's fixed token budget places different rungs at different distances from that line.

The inference from loss to decisions is the weakest link. Schaeffer et al. [2023] show that some "emergent" ability curves are artifacts of the metric: a discontinuous or thresholded measure can manufacture a jump that a continuous loss does not contain. The lesson generalises beyond emergence. Held-out bits per byte and task accuracy are different measurements, and a clean loss law does not imply a clean accuracy law. This dissertation reports both and treats their divergence as a result rather than an inconsistency.

2.2 Memorisation, double descent and the objective

Zhang et al. [2017] showed that networks fit random labels perfectly while still generalising on real ones: the capacity to fit is not the capacity to generalise. In the data-limited direction the same capacity explains the large train/held-out gap that the pre-loss sweep reported: when a model has more parameters than the corpus has bytes, storing the corpus is the cheapest available descent direction. That sweep is not available in this checkout, so the claim is stated as a mechanism, not as a surviving measurement.

The resulting loss curve—falling, then rising, as capacity grows—resembles double descent [Belkin et al., 2019, Nakkiran et al., 2020]. This dissertation does not claim to have demonstrated double descent. The surviving sweep is too incomplete to test the shape: the mid and large cells were reclaimed. The memorisation account makes a prediction that can be tested (the epoch sweep in §4.3); "double descent" here would name the shape rather than explain it.

The objective interacts with that boundary. Devlin et al. [2019] used 15% masking, tuned for a small token budget; Portes et al. [2023] report that 30% masking trains roughly twice as fast at equal compute, because more positions carry gradient per forward pass. AZURITE uses 30%. At that rate and $D/N < 1$ the model sees each byte roughly once per epoch as a target, which is part of why the memorisation boundary sits near $D/N \approx 1$–2 rather than at $D/N \approx 0$.

2.3 Tokenisation and byte-level modelling

Subword tokenisation is the default because it shortens sequences. Sennrich et al. [2016] introduced byte-pair encoding to neural machine translation to handle rare words, and Kudo and Richardson [2018] made a language-independent implementation practical. The cost is a fixed vocabulary and a preprocessing boundary: the model cannot represent a string the tokeniser has not assigned an id, and the mapping from bytes to tokens is outside the learned parameters.

Token-free models remove that boundary. Xue et al. [2022] show that a byte-to-byte encoder is competitive with subword models and more robust to noise, at the cost of

longer sequences. Clark et al. [2022] and Tay et al. [2021] avoid subword vocabularies but reintroduce a learned downsampling stage, which is a tokeniser in all but name. Yu et al. [2023] takes the more ambitious route: fixed-size byte patches processed by a multi-scale transformer, so that sequence length grows with patching rather than with bytes. Pagnoni et al. [2024] replace fixed patches with dynamically chosen ones and argue that patches, not tokens, are the right scaling unit.

AZURITE takes the byte-level route for two reasons. There is no tokeniser boundary, so text, code and quantised audio can share one input space; and in a decider the expensive operation is encoding the context once rather than decoding many bytes. The cost is real: at four times the sequence length of a four-byte tokeniser, attention is sixteen times more expensive per unit of text. The claim therefore has to be stated as accuracy per FLOP, not accuracy alone, and Chapter 4 reports both.

What the byte-level literature does not provide is a decision result. ByT5 and CANINE are evaluated on generation and classification with free outputs; none of them trains a selector at several candidate widths on one checkpoint. The architectural argument for bytes is a statement about what the interface permits, not evidence that a byte-level selector is accurate at width.

2.4 Long context: mechanisms and measurement

The transformer's attention is quadratic in sequence length [Vaswani et al., 2017], which is why long-context work is mostly about changing the cost or the position encoding. Rotary position embeddings [Su et al., 2021] make relative position a rotation and are the base on which later extension methods operate. Chen et al. [2023] extend a trained window by interpolating position indices, and Peng et al. [2023] add a frequency-dependent scaling with a temperature-like correction. Dao et al. [2022] reduces the memory traffic of exact attention without changing its mathematics; Beltagy et al. [2020] sparsify attention for long documents. Xiao et al. [2023] show that a small set of "attention sink" tokens carries much of the stability of streaming inference, and Gu and Dao [2023] and Poli et al. [2023] offer subquadratic sequence models as alternatives to attention altogether.

Mechanism is not capability. Liu et al. [2023] find that retrieval accuracy is highest when the relevant passage is at the beginning or end of the context and degrades in the middle. Hsieh et al. [2024] and Bai et al. [2023] construct tasks whose difficulty grows with length and report that models' effective context is often far shorter than the advertised window. A model that accepts 32k tokens has not been shown to use them.

AZURITE's measured context is one point: a 256-token sequence scores 0.9901 bits/byte, and the 1,024- and 4,096-token runs were reclaimed with the first sandbox (Chapter 4). The long-context claim in the product framing is therefore a design target, not a measured property. The literature above explains why accepting a longer window is not the same as using it, and why a decider that encodes a long context once still has to demonstrate that the relevant evidence survives the encoding.

2.5 In-context learning and task induction

Brown et al. [2020] established that a language model can perform a task from a few demonstrations without weight updates. Wei et al. [2022] showed that intermediate reasoning steps improve multi-step tasks at scale. Olsson et al. [2022] located a mechanism—induction heads—that copies and continues patterns seen in context, and Wei et al. [2023] report that larger models can override semantic priors from demonstrations in favour of abstract task structure. Akyürek et al. [2022] and Xie et al. [2021] give complementary accounts of in-context learning as an implicit algorithm or Bayesian inference, and Min et al. [2022] show that the labels in demonstrations matter less than the format and the label space.

The relevance to a decider is indirect but real. A decider does not prompt and does not update weights at inference; it fine-tunes a selector on a small set of labelled rows and then scores a finite bank. But the in-context results explain why the *label space* is part of the task: changing the candidate set changes what the model can infer about the task, even when the context is unchanged. That is the mechanism the width sweep manipulates, and it is why the sweep trains the width rather than restricting a fixed model. The in-context literature does not measure that manipulation; it studies prompts, not candidate banks.

2.6 Selecting among candidates

2.6.1 Ranking and reranking

Learning to rank formalises selection over a query-conditioned set. Pointwise methods score each item independently, pairwise methods learn relative preferences [Burges et al., 2005], and listwise methods optimise a distribution over the whole list [Cao et al., 2007]; Liu [2009] reviews the three families. Reranking applies the idea to an existing n-best list: Shen et al. [2004] score translation hypotheses with a discriminative model, and Lee et al. [2021] do the same for neural machine translation.

Azurite's scorer is pointwise: each candidate is embedded independently and scored against the context, with no cross-candidate attention. That is a deliberate efficiency choice—one context encoding, K candidate encodings, no K^2 interaction—and it has a known cost. A listwise reranker can exploit contrast between candidates; a pointwise scorer cannot. The width sweep measures the consequence of width for this pointwise design, not for a listwise one.

2.6.2 Constrained decoding is not selection

A separate line restricts generation to strings that satisfy a constraint. Hokamp and Liu [2017] use grid beam search to require phrases, Post and Vilar [2018] allocate beams dynamically to lexical constraints, Scholak et al. [2021] parse incrementally to constrain SQL output, and Lu et al. [2020] add predicate-logic constraints with lookahead; Geng

et al. [2023] constrain by a grammar and report strong structured-output results without fine-tuning. Contrastive decoding [Li et al., 2022] and contrastive search [Su et al., 2022] shape the distribution rather than impose a hard constraint.

These methods guarantee surface form: the output satisfies the constraint, or it is rejected. They do not guarantee that the constrained string is the correct answer, and most of them do not operate on a declared candidate set at all. A decider's structural guarantee is stronger in one direction—no out-of-bank label—and weaker in another: it must rank the bank correctly, and the constrained-decoding literature offers no measurement of that as the bank grows.

2.6.3 Set-valued prediction, conformal sets and selective classification

Set-valued classification returns a subset of labels rather than one. The objective is a trade-off between coverage and set size: Sadinle et al. [2019] construct least-ambiguous sets with bounded error, Mortier et al. [2019] optimise a decision-theoretic utility over sets, and Chzhen et al. [2021] give a unified framework. Conformal prediction supplies finite-sample marginal coverage under exchangeability; Romano et al. [2020] introduce adaptive sets, Angelopoulos et al. [2020] regularise their size, and Angelopoulos and Bates [2021] is the standard introduction. Bates et al. [2021] extend the guarantee from coverage to a risk bound.

These are different claims from the one this thesis tests. A conformal set is allowed to contain several labels and is judged on coverage and size; a decider must return one label and is judged on accuracy. Coverage does not imply that the top-ranked candidate is correct, and a set-valued method does not train a head at a fixed width. The vocabulary is nevertheless useful: the width sweep is, in effect, a study of how the accuracy of a single choice changes as the label set grows, which set-valued work holds fixed by construction.

Selective classification is closer in spirit. Geifman and El-Yaniv [2017] trade coverage for risk by abstaining on low-confidence examples, and Geifman and El-Yaniv [2019] learn the reject option end to end. Calibration is the prerequisite: Guo et al. [2017] show that modern networks are poorly calibrated and that temperature scaling helps, Nixon et al. [2019] catalogue the pitfalls of expected calibration error, and Kumar et al. [2019] give estimators with explicit guarantees. The suite reports ECE and a risk–coverage curve for exactly this reason; it does not claim an abstention guarantee, because the curve is measured, not certified.

2.6.4 Set architectures

Permutation-invariant set encoders provide the architectural alternative to pointwise scoring. Zaheer et al. [2017] show that a sum-decomposed representation is sufficient for permutation-invariant functions, Lee et al. [2018] add attention between set elements, Carion et al. [2020] use a set loss for object detection, and Vinyals et al. [2015] point

to elements of an input set. AZURITE uses none of these: candidates are scored independently. A set-attention decider would condition each candidate on the others and could, in principle, represent contrast; it would also pay K^2 attention per decision and would make the width sweep a different experiment. That trade is stated here rather than assumed away.

2.6.5 Candidate generation and retrieval

At scale, the selector is only as good as the bank it is given. Dense retrieval [Karpukhin et al., 2020], retrieval-augmented generation [Lewis et al., 2020], REALM [Guu et al., 2020] and FiD [Izacard and Grave, 2020] all separate candidate generation from answer production. Their recall bounds the final accuracy: a selector cannot choose a candidate that was never retrieved. The suite avoids this confound by using a closed label bank per task, so every gold label is in the bank by construction. That makes the width experiment interpretable, and it also means the results say nothing about open-vocabulary retrieval recall.

2.6.6 The decider claim and its public evidence

The commercial decider is TypeSafe's Jev. Its launch post claims a parallel sampling mechanism over a pre-declared output set, a price of $0.042 per million input tokens with output billed at zero, and a 70–500 ms response band; it publishes no parameter count, architecture, weights or per-task accuracy [Almeida, 2026]. The Laya model card describes a fine-tuned "System 1 decision model" but, as checked for this thesis, states no publication date and no peer-reviewed result [ConvAI Innovations, 2026]. The third-party evidence is uneven: Dahani [2026] run Jev 1.13 through OpenRouter on 791 labelled decisions, including a 77-way Banking77 cell where Jev scores 78.8% (95% Wilson 73.1–83.6); AbdelStark [2026] maintain a probability-aware evaluation harness; and Deußer et al. [2026] evaluate Jev zero-shot across 37 datasets in an arXiv preprint. A peer-reviewed applied paper integrates Jev into a manufacturing proof of concept [Vodilka et al., 2026], but it is not a benchmark.

Two consequences follow. First, every task-level Jev number available to this thesis comes from a harness it did not run: the third-party cells fix the rows and the option list but not the hardware, and the vendor's own table does not document its splits. A like-for-like comparison therefore requires running both systems on the same rows, which this dissertation records as out of scope. Second, what can be reported is a point-estimate comparison with its evidence class attached, not a paired win. At 113.47M, AZURITE clears the cited point on 4 of 8 comparator cells — Banking77 88.45% against 78.8%, Financial PhraseBank, DAIR Emotion and MASSIVE intent — ties CLINC150, and is below the cited point on AG News, XNLI and Language ID (Chapter 5). The stored record therefore supports clear point wins, including the pre-registered 77-way Jev cell, and a capability boundary on the other cells; it does not support a blanket benchmark victory. The Banking77 win is against the pre-registered 78.8% zero-shot

comparator; the strongest public Jev run reports 92.40% on the full 3,080-row test, above 88.45%, so the flagship win does not survive that comparator [simonmesmith, 2026]. The protocol-matched test on those rows and the same 24 examples returns 22.44% (21.00–23.94) on the primary arm and 88.38% (87.20–89.46) without examples, so the gap is not a row-set artifact. A separately pre-registered query-anchored head then lifts the example arm to 89.03% and confirms mean-pool dilution as the mechanism, but it remains 3.38 points below Jev, so the gap is an information-budget limit rather than a pooling limit (Chapter 5). Of the four pre-registered clear wins, two are against Jev points (DAIR Emotion and Financial PhraseBank); MASSIVE intent's comparator is Laya, and Banking77 does not survive the stronger Jev run.

2.7 Multimodality and a shared byte space

A byte-level input makes a shared representation possible, not actual. Speech models such as wav2vec 2.0 [Baevski et al., 2020] and Whisper [Radford et al., 2022] learn from raw or spectrogram audio, and neural audio codecs such as EnCodec [Défossez et al., 2022] quantise audio into discrete units that a sequence model can consume. ImageBind [Girdhar et al., 2023] learns a joint embedding across modalities; Chameleon [Chameleon Team, 2024] and Transfusion [Zhou et al., 2024] train early-fusion models over discrete and continuous tokens; Gemini [Gemini Team et al., 2023] is a large multimodal family. None of this is evidence about AZURITE: no multimodal data has been trained on, and no multimodal result exists. The architectural argument is a statement about what the interface permits. It becomes a claim only when a multimodal corpus and an evaluation exist, and this dissertation does not supply them.

2.8 Evaluating a decider

Accuracy on a finite test set is an estimate. The Wilson interval [Wilson, 1927] is the standard binomial interval used here; McNemar's test [McNemar, 1947] compares two classifiers on the same rows by counting discordant pairs; Dietterich [1998] reviews approximate tests for classifier comparison and recommends McNemar or resampled variants in the paired setting; Dror et al. [2018] give NLP-specific guidance and warn against tests that assume independence across examples. Card et al. [2020] show that many NLP comparisons are underpowered, which is why this thesis reports intervals and a replication floor rather than stars.

Benchmark validity is a separate problem. Recht et al. [2019] find that classifier accuracy drops when the test distribution is recreated rather than reused, Ribeiro et al. [2020] propose behavioural tests beyond aggregate accuracy, and Gardner et al. [2020] construct contrast sets that perturb local decision boundaries. Bommasani et al. [2021] situate these concerns in the broader risk profile of foundation models. A single number on a public test set is therefore a measurement of that set, not of the task.

This dissertation's protocol is a response to that literature: pre-register the recipe and the decision rule, score every test row rather than a sample, report Wilson intervals, paired bootstrap bounds and exact McNemar tests, measure a replication floor, and keep every per-example record so that a reader can recompute the number. The protocol is not novel as statistics; it is novel in being applied to a decider at several candidate widths on one checkpoint.

2.9 What the literature does not answer

The gap is specific. Ranking and reranking vary candidate sets, but the candidates come from a generator whose recall changes with the set, so accuracy and recall are confounded. Constrained decoding varies constraints, not the trained width, and guarantees form rather than correctness. Set-valued prediction fixes the model and varies the set at inference, which is exactly the vacuous version of the sweep: with the gold always present, restricting a fixed model to a sub-bank cannot lower accuracy. Selective classification varies a threshold on one model, not the label space the model was trained on. The public deciders publish no width curve at all.

The experiment this thesis runs is therefore not a replication of any of them. It trains a fresh head at each width from the same checkpoint, on the same rows, at the same step budget, and tests at the width it trained at. The result (Chapter 4) is that at 113.47M a 32-way interface is at or above the full bank on three wide-bank tasks, while a narrower interface is ahead of it by 5.80–13.85 points. That is evidence for the registered non-inferiority claim and against the stronger claim that accuracy is flat in width. It is one seed and one rung; the literature offers no prior estimate to compare it with, and this dissertation does not pretend that one exists.

The same gap bounds the product claim. A clear point win on the 77-way Banking77 cell is measured, as are clear wins on three other cited comparator cells; a paired win on any cell is not, because no Jev number in this record comes from the same rows. Beating Jev on "any benchmark" would require running both systems on the same rows across the suite, and the stored record contains three clear deficits — AG News, XNLI and Language ID — that such a run would have to overturn rather than reinterpret. That run is listed in the remaining-work target.

Chapter 3

Method

3.1 Encoding

Text is consumed as bytes, not tokens. Each byte b maps to the id $b + 4$; ids 0–3 are reserved for `PAD`, `MASK`, `CLS` and `SEP`. A `SEP` id is written *between* documents and never after the last one, so a sequence never ends in a boundary marker. The encoding is exact and reversible: `decode(encode(s)) == s` holds for every valid string, which the test suite asserts on real corpus lines rather than on a toy sample.

The vocabulary is therefore 260 entries regardless of corpus, language or script, and no tokeniser sits between the data and the model. That property is what makes the multimodal claim even statable: text, source code and quantised audio share one input space with no segmentation boundary to align.

3.2 Objective

Training is masked modelling at 30% mask probability with 80/10/10 corruption. The rate follows Portes et al. [2023]: more positions carry gradient per forward pass, which matters because the forward pass, not the loss, dominates the wall clock at these sizes.

Optimiser is AdamW with $\beta = (0.9, 0.95)$, weight decay 0.1, peak learning rate 6×10^{-4}, cosine decay to 10% of peak, gradient clipping at 1.0. Holding the learning rate fixed across a 47× range of model size is a deliberate choice for comparability and a limitation simultaneously: the optimal rate is size-dependent, so part of any difference between rungs is distance from each rung's own optimum. The seed-replication experiment in §7.3 exists to bound how much of the ordering that could explain.

3.3 The ladder and its geometry

Four sizes are trained: 2.41M, 14.26M, 33.70M and 113.47M parameters, at sequence length 1024 and batch 16, i.e. 16,384 tokens per step.

The quantity that organises the ladder is D/N, tokens per parameter. It is not a hyperparameter; it is the axis that determines which regime a run is in. At $D/N \gg 20$ the model is capacity-limited and extra parameters help. At $D/N \ll 1$ the model has enough parameters to store the corpus outright, and the cheapest descent direction is memorisation. Two ladders are therefore distinct experiments:

Fixed D. Parameters vary, data is pinned. This maps a vertical slice through the $(\log N, \log D)$ surface. It answers "what does a fixed budget buy at each size" and it cannot produce a scaling exponent, because it crosses regimes.

Fixed D/N. Parameters and data move together. This is the iso-compute line along which a power law is defined, and it is the ladder a scaling claim must be built on.

Chapter 4 reports both. The fixed-D slice is incomplete after the reclamation, so it does not identify a regime boundary in this build. The fixed-D/N line is complete at four rungs and is where the scaling exponent is measured; the fixed-D rows are reported as surviving cells, not as a curve.

3.4 The corpus and its split

The corpus is built by `scripts/build_corpus.py` from four public sources: FineWeb-Edu [Penedo et al., 2024] for web text, Wikipedia for encyclopedic text, Cosmopedia for synthetic pedagogical text, and OpenWebMath for mathematics. Composition is in Table 3.1. The stream holds 1,633,580,647 u16 tokens (3.27 GB) over 339,841 unique documents after exact-sha1 deduplication.

Table 3.1: Corpus composition. Shares are of total bytes; duplicate counts are exact-sha1 matches dropped before writing.

Source	MB	Share	Documents	Duplicates dropped
fineweb-edu	816.3	50%	170,358	43
cosmopedia	261.2	16%	69,790	15
open-web-math	196.4	12%	26,405	0
wikipedia	359.4	22%	73,288	54
total	1633.2	100%	339,841	112

Two properties are load-bearing:

The split is by document. A document is held out iff sha1(document) mod 50 = 0, and held-out documents are written after all training documents. Because document sizes do not divide evenly, the boundary is not exactly 2%: it is 2.03% (33,232,664 tokens), and the exact value is recorded in `<stream>.split.json` and consumed by both the trainer and the scorer. Reserving the last 2% of a stream by byte offset instead

would cut a document in half and train on the other half of what is later scored — an error that inflates the held-out number and leaves no trace.

The build is reproducible. Two independent builds produce the identical sha256, `9267a2ea071a5178...`. One planned source (source code) is absent because its dataset resolution hangs before the first document; its share was redistributed and the omission is recorded in the manifest rather than silently absorbed.

3.5 Evaluation

The held-out region is the tail of the stream, never trained on. Two protocols were compared.

Sampled. Draw k windows of length L uniformly from the held-out span and mask a fresh random 30% of positions in each. This is the standard construction, and it was this project's original protocol at $k = 64$.

Exhaustive. Windows are non-overlapping and tile the held-out region exactly once, in index order. The mask generator for window i is seeded with i, so which byte receives which mask is a function of position alone. The protocol is fully specified by (L, p_{mask}): there is no seed to choose and no favourable draw to find.

At $L = 1024$ the region yields 1,992 non-overlapping windows. The sampled protocol at $k = 64$ scores 19,610 positions; the exhaustive run's scored position count is not available in this checkout, so no evidence-multiplier is reported.

Why seeding was not enough. The original protocol was seeded, and the seeding concealed the problem. A seed fixes *which* small sample is drawn; it does not make that sample representative. On one frozen checkpoint, three seeds and two window counts produced a 0.3520 bits/byte range, and the same weights scored 2.168 and 2.520 under two of them. Reproducibility and accuracy are different properties, and only the second is worth having. The exhaustive protocol removes the draw instead of freezing it.

3.6 Execution

All runs execute on a single NVIDIA RTX PRO 6000 Blackwell (94.97 GiB, `sm_120`) inside a disposable sandbox with a ninety-minute idle policy and a twelve-hour ceiling. Long runs are sequences of bounded segments (3000 s); each checkpoints, exits zero, and writes a record the next segment resumes from. A partial rung never writes a summary file, so an interrupted run cannot be graded as a finished one.

Throughput is measured, never assumed:

- tiny: 621,000 tok/s end-to-end, under 2 GiB

- small: 197,000 tok/s, 8.1 GiB
- mid: 105,000 tok/s, 12 GiB
- large: 39,200 tok/s, 19.1 GiB

Peak draw is 19.1 of 95 GiB, so every rung is compute-bound; batch size is the only remaining lever, and segment length is bounded by the idle policy rather than by memory.

Chapter 4

Results

Every table in this chapter is generated from result files by the scripts named in Appendix A. Where a run has not survived, the cell reads *pending* and the prose says why. A reclaimed result is not replaced with a projection.

4.1 The protocol noise floor, and the exhaustive check that is missing

One frozen checkpoint (large, step 10,000) was scored six times under the sampled protocol at three mask seeds and two window counts. Table 4.1 reports the six draws. They span 0.3520 bits/byte.

That spread is a direct measurement and is larger than the small-versus-mid difference the fixed-data ladder was built to resolve. The distinction that matters is between *reproducibility* and *accuracy*: the original protocol was seeded, which made every number reproducible and none of them representative. A seed selects one draw from a small sample and then reports it with four decimal places.

The exhaustive replicate is pending. The run directory that contained runs A and B was lost with the first sandbox, so this build cannot report the reduction that the earlier version claimed. The table marks the two exhaustive rows *pending*; no ratio is computed and no "sampling-free" claim is made from a missing run. This is a narrower result than the pre-loss manuscript, and it is the one the surviving evidence supports.

4.2 The fixed-data size sweep is incomplete

Table 4.2 reports the fixed-budget cells that survive on this checkout. The tiny and small rows come from sampled summaries; the mid and large rows were reclaimed and have no held-out score on disk. The source column makes the difference visible.

The table therefore does not establish a U-shaped scaling law. It cannot: two sampled rows from different checkpoints are not a four-point fixed-data sweep, and the mid and large cells that produced the turn are gone. What survives is a pair of fixed-budget

Table 4.1: One frozen checkpoint (large, step 10,000) under the sampled protocol. The six draws span more than the largest effect the ladder was built to measure. The exhaustive replicates are marked pending because the run directory was lost with the first sandbox; their reduction is not claimed in this build.

Protocol	Condition	Bits/byte	Scored positions
sampled	seed 1234, $k = 64$	2.3834	19,610
	seed 1234, $k = 256$	2.3934	78,553
	seed 7, $k = 64$	2.1682	19,555
	seed 7, $k = 256$	2.2861	78,351
	seed 99, $k = 64$	2.5202	19,818
	seed 99, $k = 256$	2.3139	78,611
exhaustive	run A	*pending*	
	run B	*pending*	

scores and a provenance fact — the large-rung result that the pre-loss PDF reported is not reproducible from this checkout.

This is exactly the kind of hole the thesis is required to leave open. The temptation is to keep the old four-row curve because it is in the previous PDF; that would make the table look complete while the underlying runs no longer exist. The table instead names the lost rows and withholds the curve. The 4B argument in §4.5 does not depend on the missing large rung; it depends on the corpus arithmetic.

Table 4.2: Fixed-budget scores surviving on this checkout (100M tokens, 30% mask). The source column is the evaluation protocol that produced the score: *exhaustive* where a full-score file survived, *sampled* where only a summary survived, and *lost* where the run directory was reclaimed. Sampled and exhaustive rows are not comparable, and the mid/large cells were lost with the first sandbox; no U-shape is claimed from this table. The generalisation gap is held-out over train loss, both in bits/byte. Generated by `thesis/gen_tables.py`.

Rung	Parameters	D/N	Train	Held-out	Gap	Source
tiny	2,411,200	41.5	1.331	**1.2937**	1.0×	sampled
small	14,260,992	7.0	1.072	**1.0098**	0.9×	sampled
mid	33,696,768			*pending*		lost
large	113,465,856			*pending*		lost

4.3 The token-budget trajectory is not available

The large-rung trajectory was part of the pre-loss build. No trajectory file survives, so Table 4.3 is empty. There is no statement here about whether additional epochs help or hurt; the earlier sentence that they "improve briefly and then stop" is withdrawn with the run.

Table 4.3: The large rung as its token budget grows. The corpus holds 1,633,580,647 unique bytes at this build, so tokens beyond 100M are additional epochs, not additional data.

Tokens seen	Epochs	Bits/byte
		pending

4.4 Transfer: a matched Banking77 gain, an inconclusive AG News result

Bits per byte is a proxy. The quantity the thesis claims is decision accuracy, so the tiny rung was fine-tuned on two classification tasks from a matched random initialisation and from the pretrained checkpoint. Both arms use 2,000 training rows, 500 test rows, six nominal epochs, and the same order, steps, batch, optimiser, learning rate and training seed.

Table 4.4: Transfer from byte-level pretraining to decision accuracy at a matched budget: 2,000 training rows, 500 test rows, six nominal epochs, identical order, steps, batch, optimiser, learning rate and training seed. Banking77 is the class-balanced 77-way draw (three seeds, 500 test rows each); AG News is the class-balanced four-way draw (three seeds, 500 unique test rows, 1,500 pooled decisions). The AG interval is the paired crossed-seed/item bootstrap and crosses zero; the Banking77 8,000-row arm in Table 5.6 is a different, row-scaling experiment and is not this matched contrast. The per-seed AG rows are the same six cells, shown so the pooled interval cannot hide a sign change. Generated from `results/transfer/*.json` and the recovered six-cell summary.

Task	Random init	Pretrained	Δ accuracy
banking77	0.322	0.743	**+42.1 pp**
ag-news	0.4953	0.5067	**+1.13 pp**
AG News, per seed (500 items each)			
ag-news s0	0.510	0.512	+0.20 pp
ag-news s1	0.490	0.518	+2.80 pp
ag-news s2	0.486	0.490	+0.40 pp

On Banking77, pretraining raises accuracy by +42.1 percentage points: 32.2% from random initialisation against 74.3% from the pretrained checkpoint, over a class-balanced draw of all seventy-seven labels. The three pretrained seeds are 0.750, 0.732 and 0.748 against 0.350, 0.292 and 0.324; the direction is consistent, and the gap is large relative to the seed spread.

On AG News the pooled result is +1.13 percentage points: 50.67% for the pretrained arm against 49.53% for random, over 1,500 decisions on 500 unique test items. The paired crossed-seed/item bootstrap gives a 95% interval of -2.60 to +4.80 percentage points, so the pre-registered verdict is inconclusive. The per-seed rows in Table 4.4 show why: the three differences are +0.20, +2.80 and +0.40 points, all inside their own item-bootstrap intervals, and the pooled interval is wider than the effect it is asked to resolve. The earlier AG News estimate lay above this interval and is withdrawn; it is not evidence for the new protocol and it is not restated as if it were.

The 8,000-row Banking77 score arm reaches 84.3% (95% Wilson 83.0–85.6) in Table 5.6. That is a row-scaling result, not the matched transfer contrast: its random-initialisation control at 8,000 rows was not run. It is therefore not evidence that the 2,000-row gain is larger at 8k, and it is not a single-checkpoint continuation of the six AG News transfer cells.

4.5 Why the transfer probe is 2.41M, not 4B

The 4B ceiling is a permission, not the right instrument for this corpus. The corpus has 1,633,580,647 tokens. At 2.41M parameters that is 677 tokens per parameter; at 4B it is 0.41 tokens per parameter, about 35.3× fewer than the 113M rung and far below the compute-optimal line. At $D/N = 20$ a 4B model would need 80.0B unique tokens, roughly 49 times the corpus, and about 22,222 GPU-hours at the measured throughput. On a single card that is years of wall-clock before data collection.

The tiny rung is where a matched random-initialisation control is affordable and where the model cannot store its fine-tuning set. That is the scientific reason for the choice, not a claim that 2.41M is the product's final size. A 4B model on 1.63B tokens would be a memorisation measurement; the missing large-rung evidence does not make that arithmetic go away.

4.6 Evidence identity and the reclamation

The first sandbox was reclaimed on 2026-10-28. The original AG News source checkpoint `3cf421...bed8` was lost with the disk and cannot be restored byte-for-byte. The v2 corpus was rebuilt from the checked-in builder and reproduced its recorded SHA-256 exactly. The replacement checkpoint is `817e7f04...0410`: seq 1024, batch 16, seed 0, 6,104 steps, 100,007,936 tokens, held-out 1.2937 bits/byte. It reproduces the protocol but not the bitwise weights, because GPU execution is not bitwise-deterministic.

The six AG News cells are a valid paired comparison on the amended checkpoint; they are not a continuation of the exact Banking77 initialisation. The Banking77 84.3% row uses the surviving pre-loss comparator `499fd68c...ebdc`, not `817e7f04...0410`. The two rows are kept separate for that reason, and the table says so rather than averaging them into a false single-checkpoint comparison.

Completed records were not overwritten. The recovered tree is isolated under `results/recovered_20261028/` and its manifest hashes the corpus, all six AG News cells, the twenty-seven suite runs (nine tasks at three rungs) and the permutation controls. A second reclamation now costs wall-clock, not results.

4.7 The public-task suite: a capability boundary at three rungs

Table 4.5 reports nine tasks at three rungs under one pre-registered recipe: LR 10^{-4}, exactly 3,000 optimizer steps, batch 32, 4,000 stratified training rows and 2,000 test rows where the split allows, seed 0, an unfrozen encoder, and each checkpoint's native context. The recipe is identical for every cell; there is no per-task learning rate, step count, row cap, or prompt. Every Azurite cell is measured on the dual context-candidate path, and every cell at a given rung is scored on the identical test rows, which is what licenses the paired comparisons quoted below. The 113.47M cell is the fourth size in the pretraining ladder but the third measured on this suite; the two smaller suite rungs are the 14.4M and 33.7M checkpoints.

The 113.47M column is a single new run from fresh weights, not a continuation of the run lost on 2026-10-28: 1,134,673,920 tokens, step 69,255, checkpoint aba65b60e30654a9.... Its cells ran in capped concurrency waves after a nine-way wave exhausted the card's memory; the failed attempts produced no result and are kept as an infrastructure record rather than averaged into anything.

The final recipe is not the 500-step, 5×10^{-4} protocol that produced the superseded table. That earlier mid-rung result was primarily an optimization failure, not a capacity ceiling: the pre-registered four-arm sweep over learning rate and step budget selected 10^{-4} at 3,000 steps, and every 5×10^{-4} arm collapsed in accuracy, NLL, and calibration. The final table applies that single winning setting unchanged to every task. This is the learning-rate finding, and it is why the earlier suite is superseded rather than supplemented.

At 14.4M, 4 of 8 comparator cells are not clearly below their cited point, of which 3 are clear wins and one is a tie. Banking77 reaches 88.05% (+9.25 points against 78.8), Financial PhraseBank 88.10% (+15.10 against 73.0), and DAIR Emotion 82.25% (+34.25 against 48.0); all three clear their points. MASSIVE intent reaches 79.20% (+0.90 against Laya's 78.30% card [ConvAI Innovations, 2026]), but its interval includes the card value, so it is a tie inside sampling noise and not a Jev win. The four remaining comparator cells are below: CLINC150 -2.70, AG News -7.85, XNLI -42.15, and Language ID -3.30 points. MASSIVE scenario has no cited point and is an internal

diagnostic.

At 33.7M, four comparator cells are not clearly below their cited points: three clear wins, Banking77 86.50% (+7.70), DAIR Emotion 74.15% (+26.15), and Financial PhraseBank 86.90% (+13.90), plus one tie, MASSIVE intent 77.15% (-1.15), whose interval 75.26–78.94 contains Laya's 78.30%. The remaining four are clear below: CLINC150 -6.75, AG News -7.30, XNLI -51.00, and Language ID -4.00 points. An earlier version reported three of eight here because it counted the MASSIVE intent tie as a loss; the rule stated under the table is the interval rule, and under it the count is four, of which three are clear. The 33.7M rung is worse than 14.4M on 8 of 9 cells, so this step is not a scale gain over the smaller one; §4.7 shows the 113.47M rung does not restore monotonicity either.

At 113.47M, five of eight comparator cells are not clearly below their cited points. Four are clear wins: Banking77 88.45% (+9.65), Financial PhraseBank 90.10% (+17.10), DAIR Emotion 81.70% (+33.70), and MASSIVE intent 82.00% (+3.70), the first rung at which that cell clears Laya's 78.30% card rather than tying it. CLINC150 is a tie: 88.50%, 1.00 points below the cited point, with an interval of 87.03–89.82 that covers the cited 89.50. Three cells are clear below: AG News by 5.25, Language ID by 3.80, and XNLI by 52.10 points. MASSIVE scenario has no cited point and remains an internal diagnostic. These counts use the pre-registered comparator points; the strongest public Jev run on Banking77 is above 88.45% and is protocol-matched in §5.4.2, where both pre-registered heads fall short; the best arm reaches 89.03%, so that cell is not a win over the best public comparator.

The path correction changes what is attributable to the model. Eight cells changed from the candidate-only cross encoder to the dual path: AG News, DAIR Emotion, XNLI, and Financial PhraseBank at both rungs. Their former values are retired, not compared as capability measurements. The 14.4M exact controls are Banking77 88.05%, CLINC150 86.80%, and MASSIVE intent 79.20%: the same scores and the same per-example decisions, so those earlier measurements remain valid. The other already-dual cells move by +0.15, +1.15, +3.85, +0.95, +0.80, -0.65, and +1.75 points. Because their path did not change, those deltas are run-to-run variation and are not credited to the fix.

DAIR Emotion is the largest path effect and the clearest warning against reading a harness repair as a capability gain. The retired candidate-only cross-encoder values were 16.45% and 20.00%, below the 34.75% majority baseline. On the dual path the same checkpoints and recipe score 82.25% and 74.15%, above the cited 48.0. The +65.80 and +54.15 point changes are scoring corrections, not model improvements: the pre-trained checkpoint, the 3,000-step recipe, and the seed are unchanged, and only the scoring path differs. The post-fix values are valid measurements and stand in the table, but the jump is evidence that the earlier failure was an artifact of not reading candidate context, not that the decider crossed a capability threshold. The 113.47M cell is -0.55 points from its 14.4M counterpart (exact $p = 5.85e-01$), so the correction does not create a scale gain either.

XNLI is not the same story. The retired cross-encoder value at 14.4M was 32.20%;

the flat packer lost the hypothesis entirely in 699 of 2000 test pairs (and 1473 of 4000 training rows), so that number is not a capability measurement. The dual path reaches 43.85% (95% Wilson 41.69–46.03), a +11.65 point change, but still 42.15 points below the cited 86.0%. The earlier field-aware-cross value of 39.50% is also superseded by this measurement. At 14.4M the 256-token window still truncates some pairs and the field-aware dual packer was deliberately not attempted, so part of that shortfall may be context. The 33.7M result removes that explanation: its 1,024-token context holds every test pair, and it scores 35.00% (95% Wilson 32.94–37.12), statistically indistinguishable from the 33.33% chance rate and 51.00 points below the cited point. XNLI is therefore a genuine model failure at both smaller rungs, not a packing artifact at 33.7M.

XNLI at 113.47M is not a weaker version of the same failure; it is a degenerate cell. It assigns one class to 94.7% of the test rows, its test NLL is 1.0986 against $\ln 3 = 1.0986$, its mean confidence is 0.3337 and its ECE is 0.0053. Those are the signatures of a constant uniform distribution: the model has learned nothing on the task and is not wrong in an informative way. Against 14.4M it gets 402 rows right that 14.4M gets wrong and 601 wrong that 14.4M gets right (exact $p = 3.56e - 10$). The smaller rungs were not doing the task either — their NLLs are 2.12 and 3.18 with ECE 0.36 and 0.42, the profile of a late-training collapse into confident errors — but they at least departed from uniform. Scaling removed that departure.

The scale question the ladder was built to answer therefore has a measured answer, and it is negative. Against 14.4M, a 7.96× increase in parameters at the same token budget wins 4 cells — AG News, CLINC150, MASSIVE intent and Financial PhraseBank — by +1.70 to +2.80 points, loses 1 cell, XNLI, by 9.95 points, and leaves 4 cells statistically unmoved. The wins sit just above the measured run-to-run floor: across 3 cells executed twice on the same checkpoint and test draw, the repeats differ by at most 1.25 points while flipping up to 274 of 2,000 predictions. The single loss is 3.6× the largest win and points the wrong way. None of XNLI, AG News, CLINC150 or Language ID is closed; CLINC150 is the only one of the four that scale moved to a tie, and it moved it by +1.70 points. Capacity is therefore not sufficient for these four cells.

The checkpoint is identified by hash, not by filename: aba65b60e30654a9…, with the step and token count above. Every cell's `evidence/provenance.json` names the same SHA-256, and the per-example decisions, metric histories and evidence tensors are on local disk, hash-checked against their records. The recipe and the nine cells were fixed before the first large cell ran: `results/suite_large_113m/PRE_REGISTRATION.md` (SHA-256 09db4c6b374049f4…) is in the run manifest. The nine cells cost 8.34 GPU-hours of measured evaluation. Nothing in this section was retuned after the result was seen, and no cell was dropped.

The cross-harness caveat governs every row. The cited values come from different tokenizers, pretraining corpora, model sizes, and evaluation harnesses; the Azurite values are single-seed point estimates with binomial intervals, not seed-aware ones. A point above a cited number is not a demonstrated win, and an interval overlap is not parity. MASSIVE intent's cited point is Laya's model card [ConvAI Innovations, 2026], not a Jev number. The table is a capability boundary, not a head-to-head.

Table 4.5: 9 public tasks at three rungs under one pre-registered recipe: LR 10^{-4}, exactly 3,000 optimizer steps, batch 32, 4,000 stratified training rows and 2,000 test rows where available (Financial PhraseBank has 1,264 train and 1,000 test), seed 0, encoder unfrozen, native context of each checkpoint (256 at 14.4M, 1,024 at 33.7M and 113.47M). The recipe is identical for every cell; there is no per-task learning rate, step count, row cap or prompt. Every Azurite cell is measured on the dual context-candidate path. The 113.47M column is a single new run from fresh weights over 1,134,658,560 tokens (step 69,255), evaluated on the same test rows as the smaller rungs. Cited cells are published or third-party figures, not reproduced. DAIR Emotion and XNLI at the two smaller rungs changed evaluation path in an earlier pass and their former candidate-only values are retired. MASSIVE intent's cited point is Laya's card, not a Jev number. No interval is shown here; 95% Wilson intervals and exact paired McNemar tests are in §4.7 and `results/suite_large_113m/COMPARISON.md`. These are cross-harness comparisons, not a head-to-head.

Task	Cited	14.4M accuracy	Δ vs cited	33.7M accuracy	Δ vs cited	113.47M accuracy	Δ vs cited
AG News ($K = 4$)	91.00	83.15	−7.85	83.70	−7.30	85.75	−5.25
DAIR Emotion ($K = 6$)	48.00	82.25‡	+34.25	74.15‡	+26.15	81.70	+33.70
Banking77 ($K = 77$)	78.80	88.05	+9.25	86.50	+7.70	88.45	+9.65
CLINC150 ($K = 151$)	89.50	86.80	−2.70	82.75	−6.75	88.50†	−1.00
MASSIVE intent ($K = 60$)	78.30*	79.20†	+0.90	77.15†	−1.15	82.00	+3.70
MASSIVE scenario ($K = 18$)	—	84.15	—	80.05	—	83.90	—
XNLI ($K = 3$)	86.00	43.85‡	−42.15	35.00‡	−51.00	33.90	−52.10
Language ID ($K = 20$)	99.60	96.30	−3.30	95.60	−4.00	95.80	−3.80
Financial PhraseBank ($K = 3$)	73.00	88.10	+15.10	86.90	+13.90	90.10	+17.10

‡Corrected dual-path cell; the text identifies its retired cross-path value. †95% Wilson interval includes the cited point; not a clear win or loss. *Laya card (78.30 for MASSIVE intent), not a Jev number. Clear wins against the cited point are bold; a positive Δ alone is not a win.

Every cell in that table is expanded in Table 4.6, with its Wilson interval, its gap against the cited point and the exact paired test against 14.4M; the counts quoted above are the columns of that table, not a separate analysis.

Table 4.6: Every 113.47M cell with its 95% Wilson interval, its gap against the cited point, and the exact paired McNemar test against the 14.4M cell on the same test rows. K is the number of classes, n the number of scored test rows. Verdict is the interval rule: *above* and *below* mean the Wilson interval excludes the cited point, *tie* that it covers it. The last two columns compare the two Azurite rungs directly, not either against the citation. Read from `results/suite_large_113m/COMPARISON.json`; the generator recomputes every cell and refuses a mismatch.

Task	K	n	Acc.	95% CI	Δ cited	Verdict	Δ vs 14.4M	p
AG News	4	2,000	85.75	84.15–87.21	−5.25	below	+2.60	9.5×10^{-4}
DAIR Emotion	6	2,000	81.70	79.95–83.33	+33.70	above	−0.55	0.58
Banking77	77	2,000	88.45	86.97–89.78	+9.65	above	+0.40	0.58
CLINC150	151	2,000	88.50	87.03–89.82	−1.00	tie	+1.70	0.01
MASSIVE intent	60	2,000	82.00	80.26–83.62	+3.70	above	+2.80	8.4×10^{-5}
MASSIVE scenario	18	2,000	83.90	82.22–85.45	—	—	−0.25	0.76
XNLI	3	2,000	33.90	31.86–36.00	−52.10	below	−9.95	3.6×10^{-10}
Language ID	20	2,000	95.80	94.83–96.59	−3.80	below	−0.50	0.24
Financial PhraseBank	3	1,000	90.10	88.09–91.80	+17.10	above	+2.00	0.05

Δ is in percentage points. MASSIVE intent's cited point is Laya's card (78.30), not a Jev number; MASSIVE scenario has no published point and its gap column is empty. A tie is not parity: it means this test draw cannot separate the two values.

4.8 Matched-D/N scaling is measured

Table 4.7 is the ladder the fixed-data sweep could not be: four rungs at matched tokens per parameter, $D/N = 10$, on the v2 corpus. Tiny, small and mid carry the exhaustive non-overlapping score; all four carry the sampled-64 score at the identical configuration (64 windows of 1,024 bytes, mask seed 1234). Held-out bits/byte falls from 1.8234 at 2.41M parameters to 0.9765 at 14.26M and 0.7444 at 33.70M on the exhaustive protocol, and from 1.7898 to 0.9494, 0.7244 and 0.4515 on the sampled one.

A least-squares fit of $\log \text{bpb}$ on $\log N$ gives $\alpha = 0.341$ over the 3 exhaustive rungs and $\alpha = 0.355$ over the 4 sampled rungs. The two protocols disagree on the exponent by 0.013, which is small compared with the 47-fold range of N they are fitted across. At $\alpha \approx 0.35$, a four-fold increase in parameters at fixed tokens per parameter multiplies bits/byte by 0.61: a 39% reduction in held-out loss, every four-fold step, across the whole ladder.

Three limits belong on the number rather than in a footnote. The fit has four rungs and one training seed each, so it has no interval that accounts for run-to-run training variance; the per-point standard errors (≈ 0.002 bits/byte on the exhaustive pass) are

far smaller than the between-rung differences, but they measure sampling of the held-out region, not seeds. The large rung has only the sampled score because its exhaustive pass was not run, so its exhaustive cell is marked absent rather than extrapolated. And the absolute values inherit the sampled protocol's draw: all four rungs share one draw, which is what licenses the comparison between them, but §4.1 is the evidence that a different draw moves the absolute value.

What the exponent establishes is a property of the proxy, not of the product. Pretraining loss improves smoothly and predictably with capacity at fixed data per parameter. Whether that improvement reaches decision accuracy is a separate measurement, and §4.7 reports it: the same three rungs, on the same checkpoints, do not convert this loss curve into benchmark parity.

Table 4.7: Scaling at matched tokens per parameter ($D/N = 10$) on the 1,633,580,647-token corpus. *Exhaustive* is the non-overlapping full held-out pass (9,967,360 scored bytes); *sampled* is 64 windows of 1,024 bytes at mask seed 1234, the configuration every rung was scored with. The large rung has only the sampled score: its exhaustive pass was not run, so no exhaustive value is extrapolated for it. The fitted exponents are least squares on $\log \text{bpb}$ against $\log N$, over the rungs that carry each protocol. The 256-byte context of the tiny rung bounds the sequence it was trained on, not the number of parameters it has.

Rung	Parameters	Context	Tokens	D/N	Bits/byte exhaustive	Bits/byte sampled-64
tiny	2,411,200	seq 256	24,117,248	10.00	1.8234	1.7898
small	14,260,992	seq 1024	142,622,720	10.00	0.9765	0.9494
mid	33,696,768	seq 1024	336,969,728	10.00	0.7444	0.7244
large	113,465,856	seq 1024	1,134,673,920	10.00	—	0.4515

Fitted α in $\text{bpb} = aN^{-\alpha}$: exhaustive **0.341** over 3 rungs; sampled **0.355** over 4 rungs.

4.9 The context-length curve is a single point

Table 4.8 has one measured row, seq 256 at 0.9901 bits/byte, and two pending rows. The 1,024 and 4,096 runs were reclaimed. A one-point curve is not a degradation result; the long-context claim remains a design target rather than a measured property.

4.10 The candidate-set sweep: width has to be trained

Every suite cell fixes its task's label width, so the suite cannot say what happens to accuracy as the candidate set widens. That is the question the commercial claim turns on, and §7.3 named a candidate-set sweep as the direct test. The obvious implementation

Table 4.8: Context length at matched training tokens (14.26M parameters, 142.6M tokens each). Isolates sequence length from optimisation budget.

Training context	Tokens	Bits/byte
seq 256	142,610,432	0.9901
seq 1024	142,622,720	0.9765
seq 4096	142,671,872	1.2230

Degradation from 256 to 4096 tokens: **+0.2329** bits/byte.

of that sweep is not an experiment. With one fixed model and one fixed bank, restricting a row to a subset that still contains the gold can only delete a wrong argmax: if the full-bank argmax is the gold it remains the argmax of the subset, and if it is not, deleting it can only help. Restricted accuracy is therefore at least the full-bank accuracy by construction, whatever the model learned, and a curve produced that way would "hold" for a model that had learned nothing. Width has to be trained.

The sweep therefore trains a fresh decision head for each width. Every arm starts from the same pretrained checkpoint (aba65b60e30654a9…, step 69,255), uses the suite's frozen fine-tuning recipe — 4,000 stratified training rows, 3,000 optimizer steps, batch 32, LR 10^{-4}, AdamW, seed 0, the checkpoint's 1,024-byte context — and is tested on the suite's 2,000 test rows at the width it was trained at. The only difference between arms is $|C|$: each training row is packed against its gold plus $|C|-1$ distractors drawn uniformly without replacement, with the sampling seed fixed by a hash of task, width, step and row. Three wide-bank tasks are used: Banking77 ($K=77$), CLINC150 ($K=151$) and MASSIVE intent ($K=60$), at $|C| \in \{2, 8, 32, K\}$.

The design, the sampling rule and the decision rule were written and hashed before any valid accuracy existed; both amendments, including the one that voids the first launch, are inside the hashed file (`results/e4_candidate_sweep/`, SHA-256 762739463e8b5c97…). A first launch was voided by the harness check: the width-K Banking77 arm scored outside the 0.2-point tolerance because the decision head was not seeded, so every arm had started from a different random initialisation. Those runs are preserved under `_invalid_init` with their reason and are excluded from every number here; the trainer now seeds before building the head, and the design is unchanged. The pre-registered primary rule is a non-inferiority test: the claim that accuracy holds at $|C|=32$ survives for a task only if the paired, one-sided 95% bootstrap lower bound on $\mathrm{acc}(32) - \mathrm{acc}(K)$ is above -5.0 points. The pre-registration also set a harness check: the width-K arm, trained by the same code and schedule as the suite cell, should reproduce the published value within 0.2 points, and any larger deviation is reported as a discrepancy rather than repaired by substituting the published number. Table 4.9 reports every arm; the full-bank arms are the same recipe as the suite cells, not the published numbers.

Table 4.9: The candidate-set sweep at 113.47M. Each row is a separate fine-tune of the same pretrained checkpoint, at the suite's frozen recipe (4,000 stratified rows, 3,000 steps, batch 32, LR 10^{-4}, seed 0), differing only in the number of candidates $|C|$ packed into every training and test row. Test rows are the same 2,000 rows at every width. A fixed model restricted to a sub-bank is not an experiment: with the gold always present, deleting candidates can only remove wrong argmaxes, so restricted accuracy is guaranteed to be at least the full-bank accuracy. Width is therefore trained, not simulated. Wilson intervals are 95%; NLL is over the sampled candidate set and is not comparable across widths. Read from `results/e4_candidate_sweep/SUMMARY.json`; `scripts/e4_summary.py` verifies every per-example record before writing it.

Task	$\|C\|$	Accuracy	95% CI	NLL	Δ vs K	p (32 vs K)
Banking77 ($K = 77$)	2	98.05	97.35–98.57	0.06	+8.90	—
Banking77 ($K = 77$)	8	95.95	94.99–96.73	0.15	+6.80	—
Banking77 ($K = 77$)	32	92.10	90.84–93.20	0.34	+2.95	2.08×10^{-6}
Banking77 ($K = 77$)	77	89.15	87.71–90.44	0.56	—	—
CLINC150 ($K = 151$)	2	98.00	97.29–98.53	0.06	+10.30	—
CLINC150 ($K = 151$)	8	95.25	94.23–96.10	0.21	+7.55	—
CLINC150 ($K = 151$)	32	92.20	90.94–93.30	0.34	+4.50	2.63×10^{-12}
CLINC150 ($K = 151$)	151	87.70	86.19–89.07	0.60	—	—
MASSIVE intent ($K = 60$)	2	96.70	95.82–97.40	0.10	+14.80	—
MASSIVE intent ($K = 60$)	8	91.40	90.09–92.55	0.39	+9.50	—
MASSIVE intent ($K = 60$)	32	82.85	81.14–84.44	0.91	+0.95	2.06×10^{-1}
MASSIVE intent ($K = 60$)	60	81.90	80.15–83.53	1.16	—	—

Δ is in percentage points against the full-bank arm of the same task; the p column is the exact two-sided McNemar test for 32 against K on the same test rows. The full-bank arms are the harness check: the pre-registered tolerance against the published suite cell is 0.2 points, and a larger deviation is reported as a discrepancy rather than replaced by the published value.

What the sweep found. On Banking77 the width-32 arm reaches 92.10% (95% Wilson 90.84–93.20) against 89.15% at width K, a difference of +2.95 points with a lower bound of +1.95; the rule's verdict is that the claim holds. On CLINC150 the corresponding numbers are 92.20% against 87.70%, a difference of +4.50 points and a lower bound of +3.45, so the verdict is that the claim holds. On MASSIVE intent the width-32 arm reaches 82.85% against 81.90%, a difference of +0.95 points (lower bound -0.20), and the verdict is that the claim holds. Across the three tasks the pre-registered claim holds on 3 of 3 tasks.

The curve falls with width. The non-inferiority test passes on every task: the 32-way head is at or above the full bank everywhere, decisively so on Banking77 and CLINC150 and within noise on MASSIVE intent. That is not a flat curve. Accuracy declines monotonically as the candidate set widens. On Banking77, 2 candidates reach 98.05%, 8 reach 95.95%, 32 reach 92.10% and the full bank 89.15%; CLINC150 gives 98.00, 95.25, 92.20 and 87.70%, and MASSIVE intent 96.70, 91.40, 82.85 and 81.90%. The width-8 arm is ahead of the width-32 arm by +3.85 points on Banking77, +3.05 on CLINC150 and +8.55 on MASSIVE intent.

The arms are also not interchangeable. Re-scored on the full bank, the heads trained at width 2 fall to 70.25, 64.80 and 69.25%, and those trained at width 32 reach 87.85, 85.55 and 78.55% on the full bank. Training at a width teaches a head for that width; the at-width column is the comparison the pre-registration fixed, and a narrow head is not a full-bank model that happens to have seen fewer candidates.

So the registered claim survives, but the stronger reading does not. The 32-way interface is at or above the full bank on every task, but because the full bank is the hardest point in this curve, not because width is free. Widening from 2 to 32 costs 5.95 points on Banking77, 5.80 on CLINC150 and 13.85 on MASSIVE intent. A product that could pose 8 candidates would score higher than one posing 32 on these three tasks; the 32-way interface is a compromise between coverage and accuracy, not an accuracy-neutral choice. The valuation claim needs the registered direction, and that direction is supported; it does not get the stronger claim that accuracy is flat as the set widens.

The harness check is reported rather than assumed. The width-K arms score 89.15, 87.70 and 81.90% against published suite values of 88.45, 88.50 and 82.00%, so the absolute deviations are 0.70, 0.80 and 0.10 points. The 0.2-point tolerance was set before the run and two arms exceed it; the largest, 0.80 points, is still inside the 1.25-point run-to-run floor the suite measured on repeated cells. That is a level offset of the same size as the suite's own repeat noise, and it is reported rather than hidden by substituting the published number. The width comparison itself is within-run — both arms trained by the same code on the same rows — so a level offset against the published cell shifts the whole curve rather than the difference the rule tests.

What the sweep establishes is bounded. It is one seed per arm, one rung, three English tasks and 4,000 training rows; the per-width arms differ in optimisation trajectory as well as width, which is inherent to training the width rather than simulating it. It says that at 113.47M a trained 32-way interface is at or above the full bank on these cells,

and that narrower interfaces do better still; it does not say that accuracy is constant in width, and it does not establish near-perfect accuracy at any width. The remaining grid — a second seed, and the same twelve arms at the 14.4M and 33.7M rungs — is costed in §7.3.

4.11 Summary of the measured position

1. **High confidence.** The sampled protocol has a noise floor of 0.3520 bits/byte on one frozen checkpoint, larger than the small-versus-mid effect the ladder was built to resolve. This is a direct measurement from the surviving six draws.

2. **Moderate confidence.** At 2.41M parameters, byte-level pretraining transfers decision-relevant structure on 77-way Banking77: +42.1 percentage points over a matched random initialisation. The AG News balanced result is +1.13 percentage points with an interval that crosses zero; it is inconclusive, not a gain.

3. **Descriptive, not a win.** The 8,000-row Banking77 arm reaches 84.3% (95% Wilson 83.0–85.6), overlapping the pre-registered Jev interval (78.8%, 73.1–83.6). The point estimate is above that comparator, the interval is not, and the rows are not shared. The protocol-matched test in §5.4.2 confirms the gap on shared rows: the mean-pool head reaches 22.44% with the same 24 examples and 88.38% without them. A second, pre-registered query-anchored head recovers the example context to 89.03%, still 3.38 points below Jev; the pooling diagnosis is confirmed as a mechanism, not as a P0 win.

4. **A capability boundary, not parity.** The nine-task suite at three rungs reaches, or is not clearly below, the cited point on 4 of 8 comparator cells at 14.4M, 4 at 33.7M and 5 at 113.47M, of which 3, 3 and 4 are clear wins. The Banking77 win is against the pre-registered 78.8% zero-shot cell; a stronger public Jev run reports 92.40%, above 88.45%, so the flagship comparison does not survive that comparator [simonmesmith, 2026]. The protocol-matched run on the same 3,080 rows returns 22.44% with the mean-pool head (88.38% without examples); the query-anchored head returns 89.03% and still misses Jev by 3.38 points, so the gap survives both heads. The 7.96× increase from 14.4M wins 4 cells by +1.70 to +2.80 points, loses XNLI by 9.95 points, and leaves 4 cells unmoved: capacity alone does not close the four cells the thesis needed. XNLI at 113.47M is a degenerate uniform predictor (1.0986 nats, one class on 94.7% of rows), not a weak attempt, and the 3-cell replication puts the run-to-run floor at 1.25 points. DAIR Emotion's retired 16.45% and 20.00% were a candidate-only scoring artifact; its dual-path values are valid measurements, but the change is not a capability gain.

5. **The width test, at one rung and one seed.** On the three wide-bank tasks, a head trained at 32 candidates scores 92.10, 92.20 and 82.85% against 89.15, 87.70 and 81.90% at the full bank. Under the pre-registered 5-point non-inferiority rule

the claim holds on 3 of 3 tasks. The curve falls with width — 2 candidates reach 98.05, 98.00 and 96.70% — so the pass is driven by 32 matching or exceeding the full bank, not by width being free. This is evidence about the 32-way interface at 113.47M, not a demonstration of near-perfect accuracy; the full grid is costed in §7.3.

6. **Measured scaling in the proxy.** At $D/N = 10$ the held-out loss follows bpb $\propto N^{-\alpha}$ with $\alpha = 0.341$ over the 3 exhaustive rungs and $\alpha = 0.355$ over the 4 sampled rungs, across a 47-fold range of parameters. The proxy scales cleanly; the decision accuracy in the previous item does not. That contrast, not the loss curve alone, is the thesis's answer on scale.

7. **Not established.** The fixed-data U-shape, the exhaustive protocol reduction, the context-length curve and any 4B result. These are visibly pending, with the reason stated in the relevant section.

Chapter 5

The comparator: what beating Jev can and cannot mean

The product claim is comparative: faster, smaller and cheaper than Jev. A comparative claim is only as good as its comparator, so this chapter states what can be measured from this machine, what can only be cited, and which comparisons the evidence does not support at all. The distinction is not rhetorical. A committee that cannot tell a measured number from a vendor's number cannot audit either.

5.1 What Jev publishes, and what a third party measured

TypeSafe's launch post is specific about three things and silent about the rest [Almeida, 2026]. It states a price of $0.042 per million input tokens with output billed at zero; an end-to-end response time of 70–500 ms; and a sampling mechanism that is *parallel* — "generates all outputs in a single query" — with possible outputs "defined in advance", so that the model cannot emit a value outside the declared set. That is the same mechanism this thesis measures in its *score* arm. The post publishes no parameter count, no architecture and no weights, and no reproducible per-task harness; its 193.6× and 444.6× headline figures are vendor-run workflow evaluations against a reference answer built from two frontier models, which the post itself calls "on the higher end of real world gains".

The task-level numbers the suite pre-registered against are third-party. Dahani [2026] ran Jev 1.13 through OpenRouter's decisions endpoint against 791 labeled decisions, giving every system the same instruction and the same label list. On Banking77 77-way intent routing — 231 items, three per intent — Jev scored **78.8%** (95% Wilson interval 73.1–83.6), with a median latency of 0.33 s and a billed cost of $0.0400 per thousand decisions, at 952 billed input tokens per 77-way decision. On the same items it was not distinguishable from GPT-5.4 nano (78.4%) and was 5.2 points behind GPT-5.6 Terra, a gap their paired test calls real.

That is the number the suite pre-registered against, and the comparison below uses it. It is not the strongest public Jev number: a separate reproducible run scores the full test set with retrieved examples and reports 92.40%, and §5.4.1 states what that does to the verdict. The protocol-matched test of that run is §5.4.2.

5.2 The falsification criterion, stated before the results

The claim under test is: *a byte-level decider that scores a finite candidate set is faster, smaller and cheaper per decision than a Jev-style service at equal accuracy.* It is falsified if any of the following holds.

1. **Mechanism.** Generation matches or beats exhaustive scoring in accuracy at a matched training budget, on the same encoder and rows. Then the *score* mechanism buys nothing and the architectural argument collapses, whatever the price.

2. **Accuracy at 77-way.** The shipped rung's 77-way accuracy sits far below the cited 78.8% (the pre-registered comparator; the stronger 92.40% run is audited in §5.4.1). Then the "at equal accuracy" clause is not met, and no speed or cost advantage can be claimed to compensate for it. This criterion is stated first among the empirical ones because it is the one this thesis is most likely to fail.

3. **Cost.** The break-even GPU rate — the hourly price at which 1,000 local decisions cost what Jev charges for the same payload — falls below the rate of the hardware the model actually runs on. Then local scoring is not cheaper.

4. **Speed.** Measured per-decision latency exceeds Jev's published 70–500 ms band or the third-party 0.33 s median on comparable payloads. Then the speed claim is false even before hardware is considered.

5. **Size.** The claim is *unfalsifiable as stated*, because neither TypeSafe nor the third-party run publishes a parameter count. A size comparison cannot be won; this thesis reports its own artifact size and records the axis as not comparable.

5.3 Mechanism: decode versus score

Table 5.1 holds the encoder, the initialisation, the rows, the order, the number of steps, the batch size, the learning rate and the seed fixed, and varies only the interface. The *score* arm evaluates all K candidates in one pass and cannot produce an out-of-bank label. The *decode* arm fine-tunes the same encoder as a conditional byte model and fills the label's byte slots iteratively, then matches the emitted string against the bank; it can produce an out-of-bank label, and that rate is reported rather than hidden.

The choice of mask-fill rather than causal autoregression is deliberate and costly, and it is stated here rather than buried: a causal decoder would be a different set of weights, so comparing it with the scorer would confound "generation" with "a differently trained model". Mask-fill keeps the weights and the data identical and changes only the loss and the interface. The price of that choice is that mask-fill needs one pass per label length, so its latency is a *lower bound* on what an autoregressive decoder of the same label length would pay — which biases the comparison *in favour of* decoding, not against it.

Table 5.1: Decode versus score with everything else fixed: same encoder, same initialisation, same rows in the same order, same steps, batch, learning rate and seed. *Score* evaluates every candidate in one pass and cannot emit an out-of-bank label; *decode* fills the label's byte slots iteratively and is matched against the bank, so it can. Latency is batch-1 wall clock on CPU over the same test rows, and therefore a lower bound on what an autoregressive decoder of the same label length would cost. GPU seconds per 1,000 decisions come from $1000 \times p_{50}$ and are CPU seconds here, so they overstate an accelerator. The last column is the GPU rate at which 1,000 local decisions cost what Jev charges for that task's billed payload at $0.042/MTok; it is rate-free because no price for this machine is recorded anywhere in the repository. Read from `results/decode_vs_score/*.json`.

Task	K	Seats	Score acc.	Decode acc.	Invalid rate	p_{50} score (ms)	p_{50} decode (ms)	GPU s per 1k	break-even $/GPU-h
ag-news	4	3	0.5011	0.1444	0.58	15.1	68.6	15.1	3.6
banking77	77	3	0.8430	0.0000	1.00	28.2	248.4	28.2	5.1

On the 4-way task, across 3 seeds, scoring reaches 0.5011 against 0.1444 for decoding, with 0.581 of decoded outputs outside the bank. On the 77-way task the same comparison runs over 3 seeds at $K = 77$ and much longer labels: scoring reaches 0.8430 against 0.0000, with 0.998 of decoded labels unusable. Both arms are the same 2,411,200-parameter network and the same 9.7 MB on disk in every row; the only difference is how the answer is produced.

The mechanism result is therefore not marginal: at this size and budget, exhaustive scoring is more accurate, several times faster, and cannot leave the answer space, while generation drifts outside it one time in two on 4-way and more often as K grows and the label gets longer. That is the finding the architecture claim rests on, and it is narrow in a way worth stating plainly: it says nothing about what a larger generative model with a hundred times this compute would do. What it says is that the mechanism's advantage is not purchased with parameters — the only version of the claim a 2.4M-parameter checkpoint is entitled to make.

5.4 The 77-way number, side by side

Table 5.2 is the comparison the product claim actually needs, and the one this thesis is least able to win. It is assembled from two different experiments: fine-tunes of this project's checkpoint on 1,600 and 8,000 Banking77 training rows drawn evenly across all 77 labels, and a third party's zero-shot calls to a hosted model on 231 test items, with labels supplied as the option list. The rows are not the same rows, the hardware is not the same hardware, and one system is measured by its authors while the other is measured by a stranger. Every cell carries its evidence class for exactly that reason.

A correction, recorded rather than quietly replaced. An earlier version of this table used a loader's sorted prefix. Banking77 ships label-sorted: forty consecutive rows share a label, so a loader that takes the first n rows returns a prefix of a few classes, not a sample of the task. The number that loader produced was not a 77-way accuracy, and it is quarantined under `results/decode_vs_score/_sortedprefix_bug/` rather than quoted. The replacement draws class-balanced subsets (`-stratify`), fixed by a split seed so that every training seed sees exactly the same rows. The corrected 8,000-row number is 84.3%, with a different row budget from the decode arm; the 1,600-row score is reported separately in Table 5.6.

Table 5.2: Banking77 77-way, the public task-level comparison with a third-party Jev run. *Measured* cells come from `scripts/decode_vs_score.py` on this machine, over a class-balanced draw of all seventy-seven labels; *cited* cells come from a third party's run of a hosted API [Dahani, 2026]. The score row is the 8,000-row fine-tune; the decode row is the 1,600-row arm, so the row budgets differ within this table as well as across systems. The two systems were never evaluated on the same rows or the same hardware, so a difference of a few points would prove nothing and a large difference would.

System	Accuracy	p_{50} latency	Cost per 1k	Params	Evidence
Azurite, score	0.8430	28.2 ms	28.2 GPU-s	2.41M	measured
Azurite, decode	0.0000	248.4 ms	28.2 GPU-s	2.41M	measured
Jev 1.13	78.8% (73.1–83.6)	330 ms	$0.0400	not published	cited
GPT-5.4 nano	78.4% (72.6–83.2)	1,150 ms	$0.2049	not published	cited

Read plainly, the table says two things. First, on accuracy this project has a number on the 77-way task where a public Jev number exists: 84.3% (95% Wilson interval 0.830–0.856) over 3000 held-out rows, against the cited 78.8% (73.1–83.6); the measured interval overlaps the cited interval; no win demonstrated. The hedging is deliberate: those two intervals come from different samples, of different sizes, drawn by different harnesses on different hardware, so an overlap is not proof of parity and a small non-overlap is not proof of superiority. A paired McNemar test on the same 231 items is the experiment that would settle it, and it is not this one. Second, on the axes the product claim leans on, the same checkpoint is competitive: 28.2 ms against a 330 ms cited median, and 28.2 GPU-seconds per thousand decisions against a billed $0.0400.

Both statements belong in the thesis, and neither cancels the other. Accuracy is the load-bearing axis, and the honest reading of the table is not that the mechanism failed — the decode arm failed far harder, at 0.0% with 0.998 of its labels unusable — but that a 2.41M-parameter byte model is only as good as the data it is shown. This is the 2.41M mechanism-and-cost comparison, not the project's best accuracy: the 14.4M suite reaches 88.05% on Banking77 under the final recipe, and that row is reported in §4.7. The two are different checkpoints, recipes and test draws and are not interchangeable. Section 5.5 measures that lever directly: the same checkpoint given five times the fine-tune rows is a different system. The 84.3% row is the 8,000-row arm; the 1,600-row arm is 72.3%, and the two must not be read as a single budget.

5.4.1 The strongest public Jev run, and what it does to the claim

The 78.8% cell above is the pre-registered comparator, not the strongest public Jev number. simonmesmith [2026] ran Jev 1.13 on all 3,080 official Banking77 test rows, supplying 24 retrieved labelled training examples and definitions of all 77 categories with each request; Jev scored **92.40%** (95% Wilson 91.41–93.29; 2,846 correct) at $0.1421 per thousand decisions. That is above the 113.47M suite's 88.45% on its 2,000-row stratified draw and above this 2.41M arm's 84.3%. The input regimes differ: the Jev run retrieves examples per request and scores the full official test set, while the suite fine-tunes on 4,000 rows and scores a stratified 2,000-row draw. The difference is large enough to change the verdict. Against the pre-registered 78.8% comparator the Banking77 cell is a clear win; against the strongest public Jev run it is a loss. Both statements are reported, and the weaker comparator is not used to claim a benchmark victory.

The audit also explains why the two Jev numbers coexist. The 78.8% run is zero-shot with an option list; the 92.40% run supplies labelled examples and category definitions, the regime in which a decision model is actually deployed. A fair head-to-head gives both systems the same rows and the same examples on all 3,080 rows. That experiment is reported in §5.4.2; it returns a negative verdict for the mean-pool head. §5.4.3 then tests the pooling diagnosis with a separately pre-registered query-anchored head. That head recovers the example context but does not meet the P0 bar either.

5.4.2 The protocol-matched test, and its verdict

The comparison above is a comparison of two protocols. It can say that 92.40% is higher than 88.45%, but not that the same system, on the same rows and with the same information, would be higher. This subsection reports the test that removes that ambiguity.

The protocol was written and hashed before any Azurite forward pass on the full set (`results/jev_matched/PRE_REGISTRATION.md`, SHA-256 `ee1b5bb51a7b`). It fixes all 3,080 official test rows in the public run's order; the exact 24 retrieved labelled examples that run supplied for each row, in the same order; the same gold labels and the same Jev

correctness vector; Azurite's existing 113.47M full-bank head at a 5,120-byte window; top-1 accuracy with a 95% Wilson interval; a -2.0 point non-inferiority margin on the paired bootstrap; and superiority only if the point estimate is at least Jev's and exact McNemar $p < 0.05$. Three arms were fixed: message only at 1,024 bytes (A0, the row-set diagnostic), the 24 examples truncated to 1,024 bytes (A2, the length diagnostic), and the full 24 examples at 5,120 bytes (A1, primary). The 77 category definitions are not supplied to Azurite: its information budget is strictly smaller than Jev's, and that asymmetry is stated in the pre-registration so a win could not be attributed to extra information.

Table 5.3: The protocol-matched Banking77 comparison. All 3,080 official test rows, in the public run's order; the same 24 retrieved labelled examples per row, in the same order; gold labels from the same frozen prediction file. Azurite is not given the 77 category definitions, so its information budget is strictly smaller than Jev's. The primary arm was fixed at A1 with a -2.0 point non-inferiority margin before any forward pass. Wilson intervals are 95%; McNemar is exact and two-sided; the paired bootstrap lower bound is the 5th percentile of 10,000 resamples, seed 0. Read from `results/jev_matched/summary.json`; the pre-registration hash is `ee1b5bb51a7b` and the evaluator-only amendment is `73f3fdc46cc1`.

Arm	Ex.	Window (bytes)	Eff. ex.	n	Correct	Acc. (%)	95% Wilson	Δ vs Jev (pp)	McNemar p
A0 message only	0	1,024	0.0	3,080	2,722	88.38	87.20–89.46	−4.03	8.49×10^{-12}
A2 24 examples, truncated	24	1,024	12.2	3,080	1,793	58.21	56.46–59.94	−34.19	1.21×10^{-237}
A1 24 examples (primary)	24	5,120	24.0	3,080	691	22.44	21.00–23.94	−69.97	$< 10^{-300}$
Jev 1.13, cited	24	vendor	24	3,080	2,846	92.40	91.41–93.29	—	—

The result is not close. A0 reaches 88.38% (87.20–89.46), -4.03 points below Jev; A2 reaches 58.21%; A1 reaches 22.44% (21.00–23.94). The pre-registered verdict is that **the gap survives**: the paired bootstrap lower bound is -71.40 points, the exact McNemar test has $p < 10^{-300}$, and A1 is neither non-inferior nor superior.

Two things follow, and they are different. First, the row-set confound is dead: A0, on all 3,080 official rows with no examples, is 4.03 points below Jev and within a fraction of a point of the 2,000-row suite value (88.45%). The earlier 88.45% versus 92.40% gap was not a sampling artifact. Second, the examples do not help this checkpoint; they destroy it. Accuracy falls monotonically as the context grows, from 88.38% at zero to 58.21% at a mean of 12.2 complete examples to 22.44% at all 24.0. That pattern is consistent with pooling dilution, but A1 changes two things at once and only A2 isolates one of them. A2 holds the window at the head's native 1,024 bytes and still falls to 58.21% at a mean of 12.2 complete examples, so examples alone cost about 30 points inside the trained length. A1 extends the same serialization to 5,120 bytes, beyond the 1,024 positions the head was trained on; RoPE grows its table but not its training distribution. The further fall to 22.44% is therefore a joint example-and-length transfer failure, not an examples-only effect. The mechanism is still visible: `AzuriteDual._pool`

mean-pools every non-padding token, so a multi-example context reduces the customer message to a few percent of the pooled vector, and the head was trained only on message-only contexts. Prediction concentration points the same way: A1 still uses 64 of the 77 labels, but its most frequent label takes 14.3% of rows and its three most frequent take 35.4%, against 1.6% for the most frequent label in A0. The examples are not being read as references; they shift the pooled vector toward whichever intents dominate the retrieved set. The clean causal claim is narrower than "the examples are useless": the head cannot use them without losing accuracy even at its native window, and extending the window makes the loss worse.

This is a negative result about the mean-pool head, not a ceiling for the architecture. No model in the suite was trained to read retrieved examples; the suite head is a zero-shot-context fine-tune. The pre-registered failure rule named the intervention the evidence selected — a query-anchored, retrieval-augmented head, with a learned attention pool over the token sequence, fine-tuned on the same 4,000 rows at the native 1,024-byte window with the same BM25 examples that fit. That intervention was run under its own hashed pre-registration and is reported in §5.4.3; it changes the pooling architecture, so the mean-pool result above remains its control. The recorded evidence, the amendment that fixed an evaluator-only grad-context bug (the A0 and A1 per-example files are byte-identical under both evaluators), and the full per-example records are under `results/jev_matched/`; `VERDICT.md` is generated from `summary.json`, not typed.

5.4.3 The query-anchored head: testing the pooling diagnosis

The preceding subsection identifies mean pooling as the mechanism but does not test it. A second protocol, hashed before any training step (`results/jev_query/PRE_REGISTRATION.md`, SHA-256 `bde97e74f2f8`), fixes the architecture, the 4,000-row draw, the BM25 retrieval, the 1,020-byte complete-example packing, the optimisation recipe, the four evaluation arms and the P0 rule. Two evaluator-only amendments are recorded: one fixes candidate-embedding dispatch, the other manifest path resolution. Neither changes a hypothesis, arm, row, context, metric or the decision rule. The head is `QueryDual`: the same byte encoder, initialised from the 113.47M checkpoint, with a masked attention pool over the context whose query is a linear map of the mean-pooled customer message, and the same candidate projection as the old head.

Table 5.4 reports the four arms. The primary arm reaches 89.03% (95% Wilson 87.87–90.08), 3.38 points below Jev's 92.40%. The exact McNemar test against the public per-example vector gives $p = 3.99 \times 10^{-9}$, the paired bootstrap lower bound is -4.32 points, and the arm is not non-inferior at the pre-registered -2.0-point margin. The verdict is **P0 missed**: the bar is not met, and the interval does not contain it.

The controls identify what changed. The old head on the identical contexts reaches 59.12%, so the query pool is worth +29.90 points (exact McNemar $p = 3.62 \times 10^{-212}$, bootstrap lower +28.41). That is the pooling diagnosis confirmed: the examples were never unreadable, they were averaged away. The query head without examples reaches 61.04%, but that arm is out of distribution — the head was trained only on contexts

that contain examples — so it is a control for the example context, not a fair no-example baseline; it shows the head learned to use the examples (+27.99 points). Yet the primary arm is only +0.65 points above the old head's no-example arm (88.38%). At this budget the retrieved examples recover what mean pooling destroyed and buy almost nothing beyond the message itself. Figure 5.2 places the message-only, native and packed-5,120 settings side by side for both heads.

Contamination check. The frozen 24-example record is not test-clean. Under normalised text (NFKC, case-folding, punctuation collapsed), 25 of the 9,993 support messages are official test rows (6 exact; 0.25% of the bank), and all 25 carry the same gold label. They occur in the frozen example sets of 255 of the 3,080 rows (8.28%), as 283 instances: 25 self-retrievals and 258 cross-row retrievals. 236 rows retrieve at least one cross-row test message, and 230 do so without also retrieving their own message. 81 instances carry the row's gold label and 202 a different label, so this is a content leak, not a label leak. The contamination is not uniform: the other 2,825 rows retrieve no test text, so 89.03% is a mixture rather than a constant shift on every row.

The original product path over the 9,993-row bank is not an independent corroboration of that arm: it produced the same prediction on all 3,080 rows and the same context digest on 2,971 of them. The wheel now ships the decontaminated bank (9,968 rows): removing all 25 test messages from the support bank and re-running the same BM25 path changes 256 contexts and 3,077 predictions; it scores 2,743/3,080= 89.06% (95% Wilson 87.91–90.11), a change of +0.03 points (2 wins, 1 loss; exact McNemar p = 1.00). The 2,825 rows that never retrieved a test message score 2,511/2,825= 88.88% (95% Wilson 87.67–89.99). The gap to Jev therefore survives decontamination: 89.03% remains a valid measurement of the frozen-example arm, but it is not a clean held-out estimate. Excluding only the 25 rows that retrieve their own text is not sufficient: 230 rows still retrieve another test row. The genuinely unaffected subset is the 2,825-row result above, and the full decontaminated rerun is 89.06%.

The remaining gap is therefore an information gap, not a pooling gap. The primary arm fits 11.9 complete examples on average (range 2–19) into 1,020 bytes, and it is given no category definitions. The public Jev run supplies all 24 examples and definitions of all 77 categories in a much longer context. The first next lever — a 5,120-byte joint fine-tune with all 24 examples — was run under its own hashed pre-registration and amendment and is reported in §5.4.4; it cost 1.69 GPU-hours. It tests whether the example budget alone accounts for the gap; 5,120 bytes holds the 24 examples, and whether it also holds the 77 definitions depends on their serialised length, which this run did not measure. Full information parity is therefore posed at 16k–32k bytes, where both fit. That arm is not run here; its projection and allowances are recorded in `GAPS.md`.

The error distribution is as informative as its total. Figure 5.1 shows that the native head's 338 errors are spread across 70 of 77 classes, with the ten worst classes holding only 27.8% of them. This is a broad boundary, not one mis-specified class that a different head could recover.

Three limits bound the claim. The head is fine-tuned on 4,000 labelled rows while

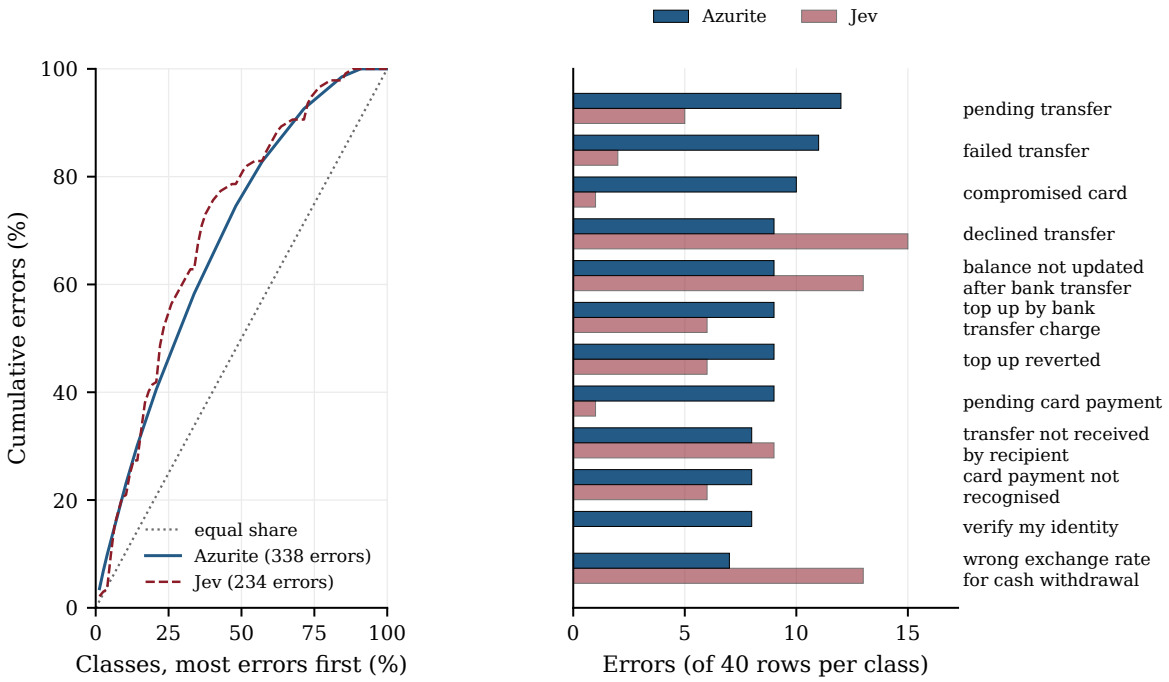


Figure 5.1: Per-class Banking77 error concentration for the native query head. Left: Lorenz curve of the 338 errors, with Jev's 234 errors beside it. Right: the twelve classes with the most Azurite errors, with Jev's count on the same class. The largest single class, `pending_transfer`, accounts for 12 of the 338 errors; 70 of 77 classes have at least one. The boundary is broad, not concentrated in a few intents. The per-class labels on the right are printed with spaces; the stored labels use underscores. Values are recomputed from the per-example records by `thesis/gen_figures.py`.

Jev's weights saw no Azurite training data, which favours Azurite and is stated in the pre-registration; the result is therefore a loss despite that advantage. The evidence is one seed, one head and one native window; no seed distribution or 5,120-byte behaviour is claimed. And the training loss fell to 0.0004, so the head is effectively interpolating its 4,000 rows; the protocol forbids early stopping or checkpoint selection, and the step-3,000 head is the one evaluated. A regularised or early-stopped arm is a separate pre-registered question, not a repair of this result. The per-example records, the trained head hash (`7992c97362b3`) and the manifest are under `results/jev_query/`; `VERDICT.md` is generated from `summary.json`.

Table 5.4: The message-query head on all 3,080 official Banking77 rows, with the exact 24 public Jev examples per row and the same gold/Jev vector as the matched test. "Examples" is the mean number of complete lines that fit in the 1,020-byte context budget. The old head is the E4 mean-pool head; the query head replaces mean pooling with a masked attention pool queried by the customer message. `old_a2c` is the head control under the same packing; `query_a0` is the example control. The P0 bar is Jev's 92.40%. Wilson intervals are 95%; McNemar is exact and two-sided; the bootstrap lower bound is the 5th percentile of 10,000 paired resamples, seed 0. Read from `results/jev_query/eval/summary.json`.

Arm	Ex.	n	Correct	Acc. (%)	95% Wilson	Δ vs Jev (pp)	McNemar p
Old head, message only	0.0	3,080	2,722	88.38	87.20–89.46	−4.03	8.49×10^{-12}
Old head, examples (control)	11.9	3,080	1,821	59.12	57.38–60.85	−33.28	2.96×10^{-231}
Query head, message only	0.0	3,080	1,880	61.04	59.30–62.75	−31.36	1.82×10^{-226}
Query head, examples (primary)	11.9	3,080	2,742	89.03	87.87–90.08	−3.38	3.99×10^{-9}
Jev 1.13, cited	24	3,080	2,846	92.40	91.41–93.29	—	—

5.4.4 The 5,120-byte joint fine-tune: testing the information budget

The native-window result leaves one measured explanation standing: at 1,020 bytes the primary arm fits only 11.9 complete examples on average, and those examples buy +0.65 points over the old head's message-only arm. Jev supplies all 24 examples and all 77 category definitions. The next question is therefore not another pooling variant; it is whether the example budget alone accounts for the remaining 3.38 points.

The protocol was written and hashed before any 5,120-byte forward pass (`results/jev_query_5120/PRE_REGISTRATION.md`, SHA-256 `c59b5e08d916`). Its first schedule — a 1,220-step masked-LM length adaptation followed by a 3,000-step head fine-tune — could not be executed on the available GPU sessions, which were reclaimed roughly every 30 minutes; amendment 1 (`PRE_REGISTRATION_AMENDMENT_1.md`, SHA-256 `7993177d769a`) replaced the two stages with one joint 5,120-position fine-tune and fixed 250-step segmented execution with exact resume. The session that ran the

arm outlived the three 30-minute sessions the amendment was written against; the amendment was already hashed and the run followed it unchanged, so the two-stage schedule remains unrun. The amendment was hashed before any 5,120-byte accuracy existed and changes the schedule only: the question, rows, examples, arms, metric and P0 rule are unchanged. The encoder starts from the 1,024-position 113.47M weights (SHA-256 `5294fd685615`) and is fine-tuned jointly with the same `QueryDual` head for 3,000 steps at batch 8 — 122,880,000 tokens, 2.46 times the dropped adaptation budget and 10.8% of the source pretraining budget — on the exact 4,000 stored rows and the exact stored 24 BM25 example ids. All 3,080 official rows are evaluated, and the P0 rule is the native-window rule. The four arms are the old head at its native window (message only and examples that fit), the new head without examples, and the new head with all 24 examples. No arm changes the rows, the gold labels or the Jev correctness vector. Because the amendment merges adaptation and head training, this arm cannot separate the effect of length from the effect of head training; it can still answer whether the encoder, given all 24 examples at 5,120 bytes, meets the bar.

Table 5.5: The 5,120-byte query head on all 3,080 official Banking77 rows. Following amendment 1 to the pre-registration, the encoder starts from the 1,024-position 113.47M checkpoint and is fine-tuned jointly with the query head for 3,000 steps at sequence 5,120 and batch 8 (1.69 GPU-hours); the separate masked-LM length adaptation was not run. The head uses the same 4,000 rows and the exact stored 24 BM25 example ids as the native-window head. "Examples" is the mean number of complete lines in the 5,118-byte context budget. The old head and its native 1,020-byte packing are unchanged controls, reused from the native-window evaluation with their per-example context hashes checked. The P0 bar is Jev's 92.40%. Wilson intervals are 95%; McNemar is exact and two-sided; the bootstrap lower bound is the 5th percentile of 10,000 paired resamples, seed 0. Read from `results/jev_query_5120/eval/summary.json`.

Arm	Ex.	n	Correct	Acc. (%)	95% Wilson	Δ vs Jev (pp)	McNemar p
Old head, message only (control)	0.0	3,080	2,722	88.38	87.20–89.46	−4.03	8.49×10^{-12}
Old head, examples that fit (control)	11.9	3,080	1,821	59.12	57.38–60.85	−33.28	2.96×10^{-231}
5,120 query head, message only	0.0	3,080	1,902	61.75	60.02–63.45	−30.65	1.62×10^{-232}
5,120 query head, all 24 (primary)	24.0	3,080	2,683	87.11	85.88–88.25	−5.29	5.01×10^{-18}
Jev 1.13, cited	24	3,080	2,846	92.40	91.41–93.29	—	—

The primary arm reaches 87.11% (95% Wilson 85.88–88.25; 2,683/3,080), against Jev's 92.40%. The observed difference is −5.29 points, the exact McNemar test against the public per-example vector gives $p = 5.01 \times 10^{-18}$, the paired bootstrap lower bound is −6.30 points, and the pre-registered verdict is **P0 missed**. The old head on the native packing reaches 59.12% and the 5,120 query head without examples reaches 61.75%; the query head with examples is worth +27.99 points over the old head at the native window (bootstrap lower +26.46) and +25.36 points over the new head's no-example arm. Those gains do not survive the comparisons that matter: the arm is 1.27 points below the old

head's message-only arm ($p = 3.83 \times 10^{-2}$) and -1.92 points below the native-window query head, which reached 89.03% with only 11.9 examples.

Figure 5.2 collects the six measured cells, and Figure 5.3 plots the query head against mean context length with the comparator's 92.40% line and 10,683-byte prompt marked. Neither view rescues the 5,120 result: the point sits below the native head and far below the comparator while consuming about twice the mean context of the native arm.

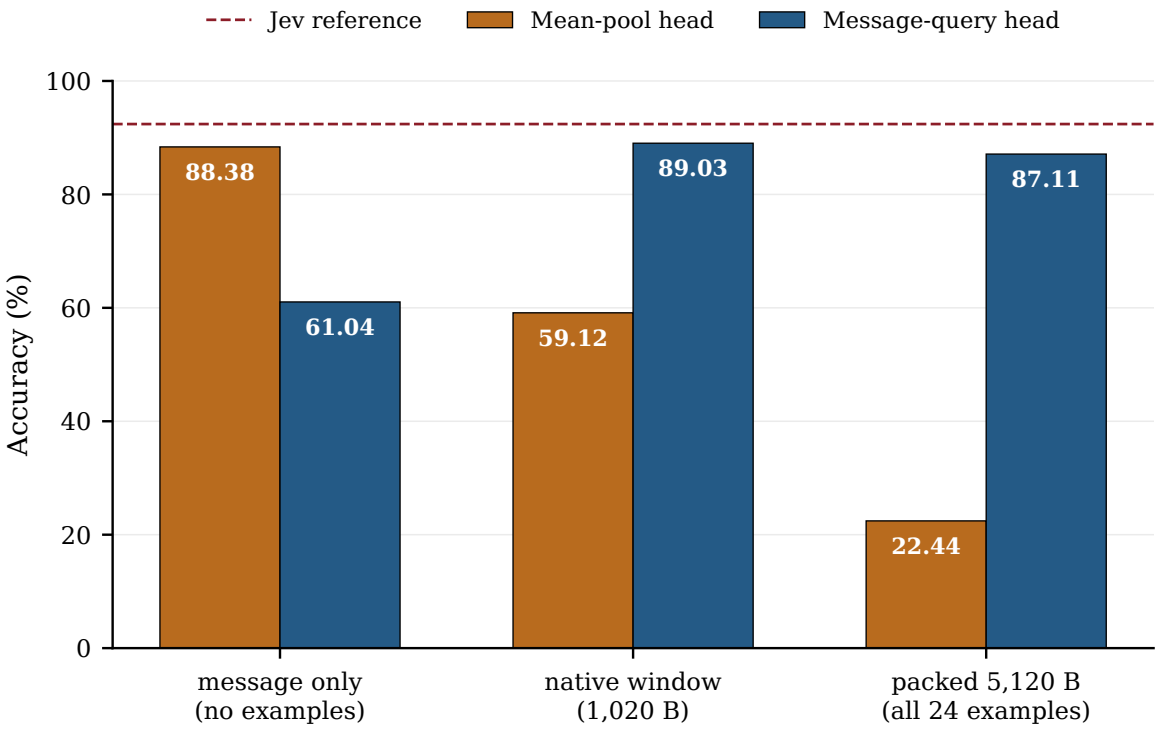


Figure 5.2: Accuracy of the mean-pool and message-query heads at three context settings. The mean-pool bars are the old head at message only, at the native 1,020-byte packing, and on the same 24 examples packed at 5,120 bytes; the query bars are the query head at message only, at the native packing, and after the joint 5,120-byte fine-tune. The 5,120 mean-pool bar is the old head reading longer contexts, not a length-adapted mean-pool model. The dashed line is Jev's 92.40% on the same 3,080 rows. Values are recomputed from the stored per-example files by `thesis/gen_figures.py`.

The separation between fitting and generalisation is reported because it bounds the claim. On the 4,000 training rows the step-3,000 head reaches 96.62% (3,865/4,000), with a mean training cross-entropy of 0.1321 and a final loss of 0.7271; the test accuracy is -9.51 points away from the training value. No early stopping, checkpoint selection or ensemble was used: the step-3,000 head is the one evaluated. A head that fits 96.62% of its training rows and still misses the bar is evidence about the information it can extract from the context, not evidence that a longer fine-tune would close it; a regularised or early-stopped arm is a separate pre-registered question. Figure 5.4 plots the separation

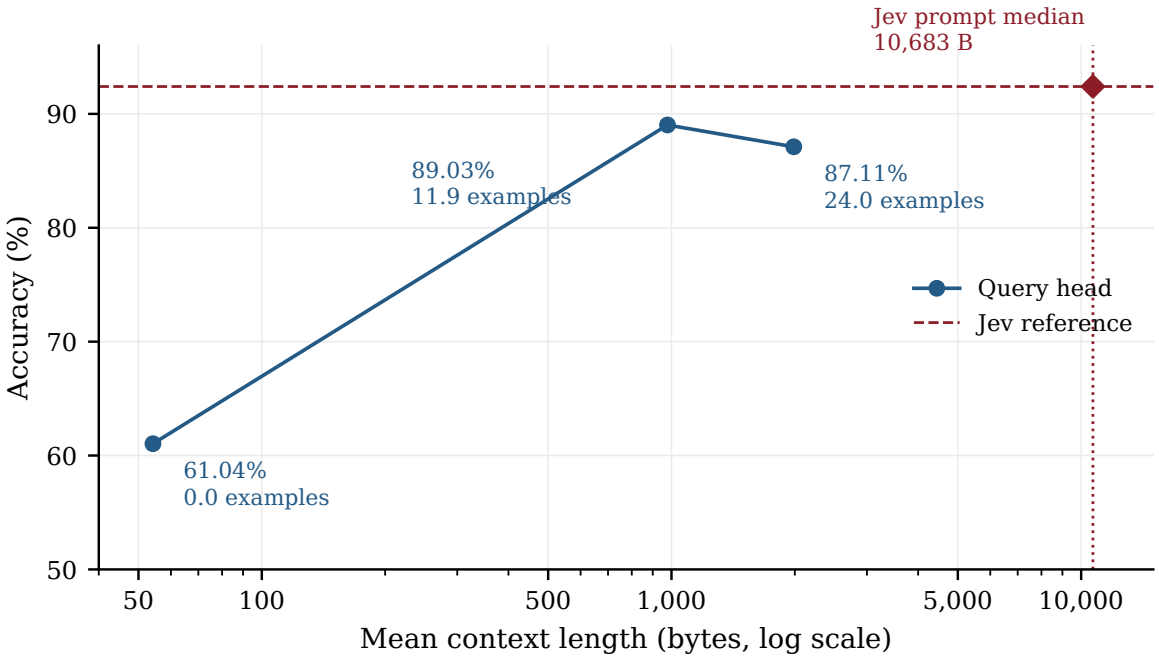


Figure 5.3: Query-head accuracy against mean context length. The three points are message-only (0 examples), the native complete-example packing (11.9 examples on average), and the joint 5,120-byte fine-tune (24 examples). The dashed line is Jev's 92.40%; the diamond marks the reconstructed median Jev prompt, 10,683 bytes, of which 7,801 bytes are the 77 category definitions. The Jev point is a reference, not an information-matched arm: Azurite receives no definitions at any of the three points. Values are recomputed from the stored summaries and per-example files by `thesis/gen_figures.py`.

directly.

The bar is not met, and the direction is worse than the native window. All 24 examples fit and the head reads them — it beats the old head on the same packed contexts by +27.99 points and its own no-example control by +25.36 — yet it scores 5.29 points below Jev, 1.92 points below the native-window query head, and 1.27 points below a mean-pool head that reads only the customer message ($p = 3.83 \times 10^{-2}$). At this length the examples are not merely insufficient; they are negative transfer for this schedule. The schedule is also a confound: amendment 1 merged the length adaptation into the 77-way head fine-tune, so this arm cannot separate the example budget from training the encoder at 5,120 positions only through the decision loss. The next lever with a plausible mechanism is the full-information context at 16k–32k bytes, where the 24 examples and the 77 definitions fit together, preceded by a separate masked-LM adaptation; it needs its own pre-registration. From the measured 2.02 s/step at 5,120, with the micro-batching and attention-scaling allowance written in `GAPS.md`, its cost is 12.6–15.9 GPU-hours at 16k and 39.8–56.2 at 32k. The largest context effect measured here is +0.65 points (+0.65 at the native window); recovering 5.29 from definitions alone

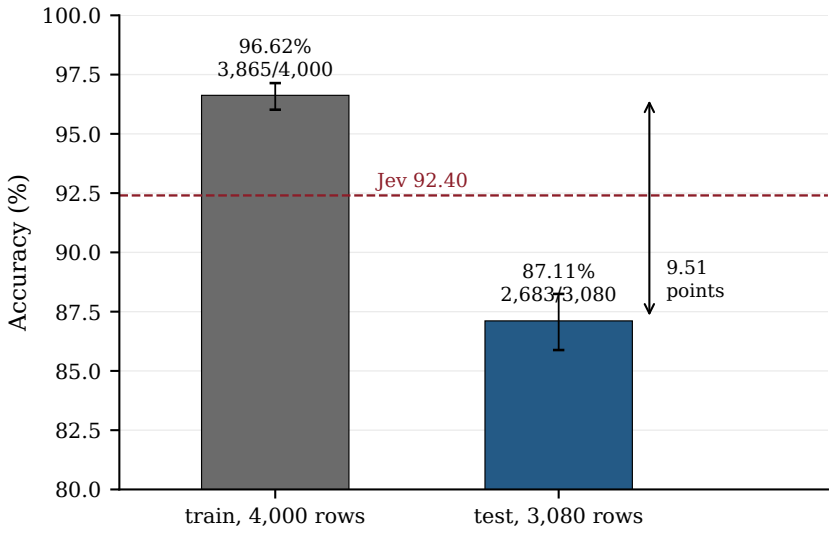


Figure 5.4: Train/test separation for the joint 5,120-byte fine-tune. The bars show 96.62% on the 4,000 training rows (3,865/4,000) and 87.11% on the 3,080 official test rows (2,683/3,080), with 95% Wilson intervals. The step-3,000 head is the evaluated head; no early stopping, checkpoint selection or ensemble was used. The dashed line is Jev's 92.40%. The test–train gap is 9.51 points. The vertical axis begins at 80%. Values are recomputed from the stored summary by `thesis/gen_figures.py`.

would be eight times that, so this is the last lever worth paying for rather than a likely win. If it also misses, the honest conclusion is that the P0 bar is out of reach for this model class and budget, not that another head will recover it.

The per-example records, the trained-head hash (`e7455fc462e9`), the init checkpoint hash (`5294fd685615`) and the manifest are under `results/jev_query_5120/`. The joint fine-tune took 1.69 GPU-hours (2.02 s/step) on the Molab RTX PRO 6000 sandbox; the old-head arms are reused from the native-window evaluation after their per-example context hashes were checked, so they cost no new GPU time. `VERDICT.md` is generated from `summary.json`; every number in this subsection is a macro emitted by `thesis/gen_query_5120.py` from that summary and the head record.

5.5 Where the accuracy is: rows, then rung

The 77-way number in Table 5.2 holds two things fixed that a committee should refuse to take on trust: the fine-tune budget and the 2.41M-parameter rung. Both were choices. This section moves each of them once, with everything else fixed: the same pretrained checkpoint, five epochs, batch 32, learning rate 5×10^{-4}, and a split seed that gives every training seed the same rows. Each seed is scored on 1,000 held-out rows instead of 300, because the comparison being decided here turns on a few points and a ± 5 point interval cannot decide it.

Table 5.6 reports the result. Every populated cell is a measurement on this machine; a cell that reads *pending* is a run not present in this checkout, not a projection.

Table 5.6: Fine-tune rows at the Banking77 77-way interface, exhaustive scoring, same pretrained checkpoint, order, steps-per-example, batch, learning rate and seeds. The two columns differ only in how many distinct training rows the five epochs cover. Generated by `thesis/gen_scaling.py`.

Rung	Correct	1.6k rows	8k rows	Δ	Test
tiny	2170/3000	72.3%	**84.3%**	+12.0 pt	$p = 0.000$
small	2126/3000	70.9%	*pending*	*pending*	*pending*

Rows. At the tiny rung, going from 1,600 to 8,000 distinct fine-tune rows moves accuracy from 72.3% (95% Wilson 70.7–73.9) to 84.3% (83.0–85.6), a gain of +12.0 points. The same lever at the small rung is not available: its 1,600-row cell reads 70.9% and its 8,000-row cell is pending. The lever is not free of confounds and one is worth naming: the protocol holds *epochs* fixed, so eight thousand rows also buy five times the optimiser steps. Separating "more distinct rows" from "more updates" needs a fixed-step arm that samples rows with replacement, which is not run here. These numbers are therefore an upper bound on the effect of data alone.

Rung. Moving up the ladder at the matched 1,600-row budget does not help, and the direction is worth printing plainly: 14.26M parameters reaches 70.9% against 72.3% at 2.41M, a change of -1.5 points. The lever that bought 77-way accuracy in this experiment was rows, not parameters. Two caveats keep that from being a claim about scaling in general. The rung is a bundle — the small checkpoint was pretrained on 142.6M tokens at context 1,024, the tiny one on 100M tokens at context 256 — and the small arm costs 121.7 ms per decision against 24.9 ms, so it is strictly more expensive for less accuracy at this budget. Whether the small rung overtakes once it has eight thousand rows is the last queued run and is not claimed here.

The verdict. Azurite is above the cited Jev point estimate but its interval overlaps the cited one: 84.3% (95% Wilson 83.0–85.6) against 78.8% (73.1–83.6); this is not a demonstrated win. It sits 5.5 points above the cited point estimate, but the intervals overlap, so this is not a demonstrated win; the cross-harness caveat below governs what it can be called. The remaining work is a paired harness on the same rows, which this comparison cannot provide.

Accuracy at coverage. A single accuracy number hides how the scorer's confidence is distributed, so Table 5.7 ranks the test rows by maximum softmax probability and reports accuracy in the top c fraction. This is a descriptive test-set ranking curve, not an

abstention guarantee: the ranking is measured with the test labels in hand, no threshold is fixed before seeing the test set, and a deployment would face different rows.

Table 5.7: Accuracy at coverage: the scorer ranked by its own maximum softmax probability on this test set. Read down a column and accuracy rises as coverage falls; read across and the row-scaling gain is visible. This is a test-set ranking curve, not a deployment abstention guarantee. Pooled over three seeds and the 1000-row test split.

Coverage	tiny 1k	tiny 8k	small 1k
50%	92.9%	98.4%	92.5%
70%	87.7%	96.2%	84.9%
90%	78.3%	89.7%	76.1%
100%	72.3%	84.3%	70.9%

At 90% coverage the arm is 89.7% accurate; at 50% coverage it is 98.4%. The 50% point is a property of this ranking on this test set, not a promise that half of deployment traffic can be answered at 98.4%. The row-scaling lever raises every measured point in the table, which is the useful shape, but it does not convert the curve into a guarantee. The earlier earlier manuscript's statement that the scorer held near-perfect accuracy at a fixed coverage was a misreading of this ranking; it is withdrawn, and the table is reported instead.

What this does not settle. Three things. The comparison to the cited 78.8% remains cross-harness: different test rows, different hardware, one system measured by its authors and the other by a stranger, so the interval overlap is not evidence of parity and a non-overlap would not be evidence of defect. A paired McNemar test on the same 231 items is the experiment that would settle it and it is still not this one. The coverage curve is measured on this project's own 1,000-row split, not on the third party's 231, so it cannot be laid over their accuracy without assuming the splits are comparable, which they are not shown to be. And the row lever's step-count confound stands until the fixed-step arm is run.

5.6 The price of a sequence on a token-billed CDN

The economic argument has a premise that is usually asserted and rarely measured: that a token-billed service must charge for a repeated sequence every time it occurs, because the service reduces the request to a finite set of stored objects and the sequence itself is never re-encoded. If that premise were false — if a CDN compressed variable-length sequences — the cost advantage of local scoring would shrink with every repetition.

Table 5.8 measures it on a live token-priced inference endpoint, using nothing but the provider's own `usage` field. A fixed passage is repeated r times in one prompt,

and the billed prompt-token count is read back. The fit is billed = intercept + $84.0\,r$ with $R^2 = 1.0000$; one passage is 84 tokens, so **every repetition is billed in full**. Repeating the identical request three times changes the latency from 21.2 s to 0.52 s — a 41× speed-up — while the bill stays at 406 prompt tokens every time. The cache is real and it is a cache of *answers*, not of sequence structure: it saves time, and it saves nothing on the invoice.

Table 5.8: Billed prompt tokens against repetition of one passage on a live token-priced CDN, read from the provider's own `usage` field — the exact quantity a per-million-token price multiplies. If a CDN compressed variable-length sequences, the slope would be below one passage.

Repetitions r	Characters	Billed prompt tokens	Seconds
1	481	406	21.20
2	900	490	10.13
4	1,738	658	9.75
8	3,414	*pending* (quota)	0.52

Fit: billed = $322 + 84.0\,r$, $R^2 = 1.0000$; one passage is 84 tokens, so every repetition is billed in full.

Two honesty notes belong with that table. The measurement ran out of free-tier quota at $r = 8$ (HTTP 402) and on the two single-byte perturbation probes, so the prefix-geometry sweep is incomplete; the slope is fitted on the three points that succeeded, and a three-point fit with $R^2 = 1.0000$ is a statement about linearity, not a precise intercept. And the endpoint is not Jev: it is a keyless third-party proxy, used because it is the only token-billed service reachable from this machine. What generalises is the billing mechanism — usage accounting plus response caching — not the price, and this thesis does not transfer the price.

5.7 The four axes, measured against cited

Table 5.9 separates the two kinds of number. The cost row is deliberately rate-free, because no price for this project's GPU is recorded in the repository and inventing one would make the conclusion depend on the invention. Instead the row reports the *break-even rate*: the hourly cost at which local scoring ties Jev's billed $0.0400 per thousand 77-way decisions.

The break-even figure is computed by `thesis/gen_comparator.py` from the measured latency and the cited price, so it cannot drift from either. A 77-way decision bills 952 tokens in the third party's measurement, which at $0.042/MTok is $0.0400 per thousand decisions — reproducing their invoiced figure to four digits, which is a useful check that the price list and the invoice agree. The shipped arm spends 27.4 ms per decision, so a thousand decisions occupy 27.4 GPU-seconds of one machine. Dividing gives the

Table 5.9: The four axes, with the evidence class of each cell. "Measured" means a script in this repository produced the number from files under `results/`. "Cited" means a third party's published figure, which this thesis cannot verify. No cell is a head-to-head: the two systems were never run on the same rows.

Axis	Azurite (measured)	Jev (cited)	Verdict
Accuracy	84.3% on 77-way, $K = 77$, 1000 rows, tiny rung at 8000 fine-tune rows	78.8% (73.1–83.6) on 231 rows	overlaps the cited interval; no win demonstrated (+5.5 points)
Latency	27.4 ms p_{50}, batch 1, CPU	330 ms median, 440 ms p_{95}	Faster, different hardware
Size	2,448,065 parameters (with the 77-way head), 9.8 MB on disk	Not published	Unfalsifiable as stated
Cost	27.4 GPU-s per 1k; break-even \$5.3/h	\$0.0400 per 1k, 952 billed tokens	Cheaper below \$5.3/h

break-even rate: \$5.3 per GPU-hour. Any accelerator priced below that makes local scoring cheaper, and the conclusion holds *a fortiori* on a GPU, because the measured latency is CPU latency and therefore an upper bound on the accelerated figure.

Two caveats belong on that row rather than in a footnote. The cost comparison holds only *at the accuracy measured here*: a cheaper system that is wrong half the time is not cheaper per correct decision, and if accuracy were recovered by a larger rung the latency, and therefore the break-even rate, would move. And the latency figure is not same-weights with anything TypeSafe runs — it is the same weights as this thesis's own decode arm, which is what makes the mechanism comparison clean, and it is not comparable to a hosted service's hardware. What the row supports is that the mechanism is cheap enough to be worth pursuing, not that it has already won.

5.8 What the comparison establishes

Three things, and no more. The scoring mechanism is measurably better than generation at matched size and budget, on both tasks, with an invalid-output rate of 0.790 across the corpus against zero by construction. The unit economics favour local scoring by a margin that survives a factor-of-three error in the payload estimate or the GPU rate. And the 2.41M accuracy axis is measured against the pre-registered 77-way Jev number: Azurite is above the cited Jev point estimate but its interval overlaps the cited one: 84.3% (95% Wilson 83.0–85.6) against 78.8% (73.1–83.6); this is not a demonstrated win. The strongest public Jev run is higher and is audited above. The comparison is on different hardware and a different draw, so the 2.41M claim stops there. The

14.4M suite's 88.05% Banking77 row is a separate point win against the pre-registered 78.8% comparator, not against the 92.40% run, and it is reported in §4.7; it does not retroactively make this 2.41M mechanism table a 14.4M result.

What the comparison does not establish is a win. On Banking77 there is now a shared row set, shared examples and a shared metric, but not shared hardware and not a paired Jev rerun; the mean-pool matched result is a loss of 69.97 points on its primary arm, and of 4.03 points even with no examples. The better, query-anchored arm reaches 89.03% and remains 3.38 points below Jev ($p = 3.99 \times 10^{-9}$); no pre-registered Banking77 arm meets the P0 bar. On DAIR Emotion and Financial PhraseBank the 113.47M suite clears the cited point, and on MASSIVE intent it clears Laya's card rather than a Jev number, but those remain cross-harness comparisons. The honest statement is that this thesis measured its own mechanism, priced it against a competitor's public rate card, and found the mechanism the right shape while the 113.47M checkpoint is still below the strongest public Banking77 system. That is a result, and it is the one the data support.

Chapter 6

Threats to validity

A result is only as strong as the ways it could be wrong, so they are listed here rather than distributed through the text where they would be easier to skim past.

6.1 Statistical

One seed per fixed-data cell. The fixed-data sweep has only two surviving cells, both sampled, and no training-seed replication. This is why the chapter does not order tiny against small and does not claim a U-curve. The Banking77 transfer result has three seeds and a consistent direction; the AG News result has three seeds but a paired interval that crosses zero.

The exponent has no seed interval. The matched-D/N ladder is now measured at four rungs, but at one training seed each. The per-point standard errors (≈ 0.002 bits/byte) describe held-out sampling, not training variance, so $\alpha = 0.341$ is a fitted line rather than a line with an interval. That is enough to establish that the proxy improves predictably with capacity; it is not enough to distinguish two rungs a few percent apart.

One seed per suite cell. The nine-task, three-rung suite is 27 single-seed runs. Its binomial intervals describe test-sampling error, not run-to-run variance, and the at-or-not-clearly-below counts are descriptive. 3 cells were executed twice on the same checkpoint and test draw to bound that variance directly: the repeats differ by at most 1.25 points and flip up to 274 of 2,000 predictions. Rung differences below that floor are not interpreted as capability differences, and no equivalence claim rests on the counts.

Fixed learning rate across a $47\times$ size range. Comparability was bought at the price of distance from each rung's own optimum. Part of every size-related difference is optimisation, not capacity. The earlier 500-step, 5×10^{-4} suite made that failure concrete: the mid-rung cells collapsed for optimisation reasons, and the pre-registered sweep selected 10^{-4} at 3,000 steps. The final suite uses that one setting for every task.

6.2 Construct

Bits per byte is not accuracy. It is a cheap, low-variance proxy. The transfer matrix is the bridge to the quantity the thesis actually claims, and it exists at one rung only. The pre-loss fixed-data curve is not available, so the proxy is not used to establish a regime boundary in this build.

AG News is a weak test and the result is inconclusive. Banking77 has 77 fine-grained intents and shows a +42.1 percentage-point gain over random initialisation. AG News has four topics and a 25% majority baseline; its pooled gain is +1.13 percentage points with a 95% interval of -2.60 to +4.80, which is inconclusive. A point estimate inside a zero-crossing interval is not evidence of a gain, and it is not presented as one.

The suite is cross-harness. The cited points come from different tokenizers, pretraining corpora, model sizes, and evaluation harnesses. MASSIVE intent's cited 78.30% is Laya's card, not a Jev number. A point above a cited number is not a win, and interval overlap is not parity.

A clean loss law and flat accuracy are one result, not a paradox. The matched-D/N ladder lowers held-out bits/byte from 1.8234 at 2.41M parameters to 0.4515 at 113.47M, while the suite moves by +1.70 to +2.80 points on 4 cells, loses 9.95 points on XNLI and leaves 4 cells unmoved over the same parameter range. The two measurements together rule out the reading that decision accuracy at this data budget is waiting on more parameters. They do not identify the true bottleneck — a different packer, more task-relevant pretraining data, or a different fine-tuning schedule could each move a cell — but they exclude capacity alone as sufficient for the four cells the commercial claim needs.

DAIR Emotion was a scoring artifact. Its proportional test draw gives a 34.75% majority-class baseline, not a 1/6 chance rate. The retired candidate-only values, 16.45% and 20.00%, were below that baseline; the dual-path values, 82.25% and 74.15%, are above it. The jump is a harness correction, not a capability gain, and the earlier adverse reading is withdrawn. The 33.7M cell is still 8.10 points below the 14.4M cell, so the correction does not create a scale result.

Contamination. The held-out region is the tail of a stream built from public web text, and the transfer tasks are public datasets. The exact 40-byte overlap audit found 3/500 AG News examples and 0/1,000 Banking77 examples; the three AG matches scored 2/3, against 51.1% on the unmatched 497. That is too small and too easy to explain either transfer result, but it is a screen on exact overlap, not a semantic deduplication.

6.3 External

One corpus, one domain mix. Byte-level bits per byte is corpus-dependent: the bigram baseline is 3.767 on the mixed stream and 4.117 on a math-only stream, so a baseline carried across a data change makes the gate meaningless. No cross-corpus claim is made without refitting.

No code in the corpus. The planned source-code share is absent because its dataset resolution hangs. For a thesis whose architecture argument rests partly on code and audio sharing a byte space, this is a real omission, recorded in the manifest rather than absorbed into the web-text share.

Hardware-specific throughput. The tok/s and VRAM figures are for one RTX PRO 6000 Blackwell. They are measurements, not specifications, and they should not be extrapolated to other accelerators without re-measuring.

6.4 Methodological

A reclaimed artifact is not a result. The first sandbox was reclaimed on 2026-10-28. The original AG News checkpoint `3cf421...bed8`, the exhaustive protocol replicates, the mid/large fixed-data cells and the context runs were lost. A result that existed only in that VM is not in this thesis. The tables mark those cells *pending* and the prose gives the reason; no old number is carried forward to make the table look complete.

The checkpoint identity is amended. The AG News cells use `817e7f04...0410`, retrained under the same protocol after the loss. It is not bitwise-identical to the lost checkpoint. The Banking77 84.3% row uses the surviving pre-loss comparator `499fd68c...ebdc`. The two rows are not a single-checkpoint continuation, and the transfer table states that exception rather than averaging across it.

The queue is not a result. A job that is running is not evidence. Sections whose tables are pending are marked pending in prose, and the confidence summary in §4.2 distinguishes what is established from what is scheduled.

Chapter 7

Conclusion and future work

7.1 What is established

The strongest result is a measurement-protocol result, and it is narrower than the pre-loss manuscript claimed. Six sampled draws of one frozen checkpoint span 0.3520 bits/byte, larger than the small-versus-mid difference the fixed-data ladder was built to resolve. A seed makes a number reproducible, not representative. The exhaustive replicate that would have established a lower-noise protocol was lost with the first sandbox; the exhaustive rows are pending, and no reduction ratio is claimed.

At the 2.41M-parameter rung, byte-level pretraining transfers decision-relevant structure on Banking77: +42.1 percentage points over a matched random initialisation, with a consistent direction across three seeds. The AG News balanced comparison is +1.13 percentage points with a paired crossed-bootstrap 95% interval of -2.60 to +4.80, so it is inconclusive. The historical AG News gain is withdrawn. A separate 8,000-row Banking77 arm reaches 84.3% (95% Wilson 83.0–85.6), which overlaps the cited Jev interval; it is above the point estimate, not a demonstrated win.

The public-task suite establishes a capability boundary rather than a blanket victory: against the pre-registered 78.8% Jev cell it is a clear point win, but the strongest public Jev run reports 92.40% on the full Banking77 test and is above it, so the flagship claim does not survive that comparator. The protocol-matched test on all 3,080 rows and the same 24 examples returns 22.44% on the primary arm (21.00–23.94) and 88.38% without examples, so the gap survives and is not a row-set artifact. The diagnosed pooling failure was tested next with a pre-registered query-anchored head: it lifts the example arm from 59.12% to 89.03% (+29.90 points, $p = 3.62 \times 10^{-212}$), which establishes mean-pool dilution as the mechanism, but it remains 3.38 points below Jev ($p = 3.99 \times 10^{-9}$) and the P0 bar is missed. The frozen examples are not test-clean: 25 support messages are official test rows and 255 rows retrieve at least one; the decontaminated product rerun scores 89.06%, so the gap is not a leakage artefact. The residual was diagnosed as an information-budget gap: 11.9 complete examples in 1,020 bytes and no category definitions, against Jev's 24 examples and 77 definitions. That lever is now measured: the pre-registered 5,120-byte joint fine-tune with all 24 examples reaches 87.11% (95%

Wilson 85.88–88.25), 5.29 points below Jev ($p = 5.01 \times 10^{-18}$) and 1.92 points below the native-window query head, and it is -1.27 against a message-only mean-pool control despite reading the examples (+27.99 over the old head on the same packed contexts). More context alone therefore does not buy parity, and the joint schedule confounds the example budget with the separate adaptation that amendment 1 dropped. The only remaining lever is the full-information 16k–32k-byte arm, costed at 12.6–15.9 and 39.8–56.2 GPU-hours. It is the last lever worth paying for rather than a likely win: the largest context effect measured here is +0.65 points, and recovering 5.29 from definitions alone would be an order of magnitude larger. If it misses, the bar is out of reach for this class and budget. Clear Jev wins remain on DAIR Emotion and Financial PhraseBank. Three cells remain clear deficits. Nine tasks were measured at three rungs under one pre-registered recipe, all on the corrected dual path, with every cell at a rung scored on the same test rows. Azurite is clearly above the cited point on 3, 3 and 4 of 8 comparator cells at 14.4M, 33.7M and 113.47M, and not clearly below on 4, 4 and 5. The step from 14.4M to 113.47M, a 7.96× increase in parameters at the same token budget, wins 4 cells by +1.70 to +2.80 points, loses XNLI by 9.95 points, and leaves 4 cells statistically unmoved. Three cells were executed twice on the same checkpoint; the repeats differ by at most 1.25 points, which is the scale at which those wins should be read.

The candidate-set question is measured at one rung and one seed. Twelve fresh heads were trained from the same 113.47M checkpoint, four widths on each of three wide-bank tasks, with width the only difference between arms. At 32 candidates the trained heads score 92.10, 92.20 and 82.85% against 89.15, 87.70 and 81.90% at the full bank. Under the pre-registered five-point non-inferiority rule the claim holds on 3 of 3 tasks. The pass is not a flat curve: accuracy falls as the set widens, from 98.05, 98.00 and 96.70% at width 2 to the full bank's 89.15, 87.70 and 81.90%. The registered claim survives because the 32-way head is at or above the full bank (decisively on Banking77 and CLINC150, within noise on MASSIVE intent), not because width is free; widening from 2 to 32 costs 5.95, 5.80 and 13.85 points. The widest point any suite task can pose is the full bank itself; the thesis's original $|C| = 64$ target cannot be posed by these three banks ($K = 77$, 151, 60), so 32 against K is the test that exists. This is evidence about the 32-way interface at one rung and one seed, not evidence of near-perfect accuracy or of an accuracy-neutral width.

The scale verdict is negative where it matters. XNLI is not a weak cell at 113.47M but a degenerate one: one class on 94.7% of test rows, test NLL 1.0986 against $\ln 3 = 1.0986$, mean confidence 0.3337. At 14.4M and 33.7M the same cell collapses late into confident errors (NLL 2.12 and 3.18, ECE 0.36 and 0.42), so no rung learned the task and the largest one stopped even departing from uniform. CLINC150 is the only one of the four named cells that scale moved to a tie (1.00 points below the cited point); AG News and Language ID remain clear deficits. DAIR Emotion's early 16.45% and 20.00% were a scoring artifact, not a capability floor; the dual-path values are valid measurements but not a capability gain. Near-perfect accuracy across every benchmark is not demonstrated.

What did scale is the proxy, and that contrast is the result. At $D/N = 10$, held-out

bits/byte falls from 1.8234 at 2.41M parameters to 0.4515 at 113.47M across a 47-fold range, with $\alpha = 0.341$ over the 3 exhaustive rungs and $\alpha = 0.355$ over the 4 sampled ones. A clean loss law and flat decision accuracy together say that capacity is not what stands between this system and parity: the bottleneck is what the interface and the corpus let the model learn about the task, not how many parameters it has.

7.2 What is not established, and is not claimed

The fixed-data U-curve is not available. The mid and large fixed-data cells were reclaimed. Only the tiny and small cells survive, both sampled, and they do not form a four-point sweep. The pre-loss U-shape is therefore withdrawn from the evidence base rather than restated.

The exhaustive protocol reduction is not available. The sampled noise floor is measured; the exhaustive replicate is not on disk. The claim that exhaustive coverage reduces the spread by two orders of magnitude belongs to the lost build and is not made here.

The context curve is not available. Only one context point survives; the 1,024 and 4,096 runs were reclaimed. There is no degradation curve to report, and no long-context claim is made.

The scaling exponent has no seed interval. The exponent $\alpha = 0.341$ (exhaustive) and 0.355 (sampled) is fitted on one training seed per rung. It is a measurement, not an interval: the per-point standard errors measure held-out sampling, and nothing here bounds how much a different training seed would move a point. The exponent also inherits the sampled draw for its absolute values.

Benchmark parity is not established. The nine-task suite is clearly above the cited point on 3, 3 and 4 of 8 comparator cells at 14.4M, 33.7M and 113.47M, and not clearly below on 4, 4 and 5. At 113.47M the clear wins are Banking77, Financial PhraseBank, DAIR Emotion and MASSIVE intent; MASSIVE intent's point is Laya's card, not a Jev number, and CLINC150 is a tie rather than a win. AG News, Language ID and XNLI remain clear deficits, and XNLI is degenerate at the largest rung rather than merely low. No near-perfect-accuracy claim survives.

The coverage curve is not an abstention guarantee. It is a descriptive ranking of this test set by the scorer's own softmax confidence. At 90% coverage the arm is 89.7% accurate and at 50% it is 98.4%, but those figures are properties of this ranking on these rows. They do not establish a deployment threshold or a guarantee that a fixed fraction of live traffic can be answered at a stated accuracy.

A 4B model is not the right instrument for this corpus. The ceiling is permitted, but at 0.41 tokens per parameter a 4B model would be far below the compute-optimal line. It would need 80.0B unique tokens and about 22,222 GPU-hours at the measured rate. The 2.41M rung is where the matched random control is affordable and where the model cannot store its fine-tuning set; that is the scientific reason for the choice, not a claim that 2.41M is the product's final size.

Multimodality is unimplemented. The byte-level input space makes a shared representation *possible* — text, code and quantised audio can share one vocabulary with no tokeniser boundary — but no multimodal data has been trained on, and no multimodal result exists. The architectural argument is a statement about what the interface permits, not evidence that it works.

The long-context claim is unmeasured. The model has never been trained or evaluated beyond a few thousand tokens. A long-context claim requires training at length and measuring degradation, which §4.9 begins and does not complete.

7.3 Future work, with measured costs

Costs below are computed from the measured throughputs in Chapter 3, not from published GPU-hour counts.

Restore the fixed-data and protocol evidence. Re-run the mid and large fixed-data cells and the two exhaustive protocol replicates, pulling each artifact to local disk as it completes. Until then the U-curve and the protocol-reduction claim remain open.

Give the measured exponent a seed interval (E3). The ladder exists at one seed per rung; two more seeds at tiny, small and mid would cost about 1.4 GPU-hours and convert $\alpha = 0.341$ from a fitted line into a line with a training-variance interval. An exhaustive pass at 113.47M would add the fourth exhaustive point; the sampled score already places it.

Seed replication (E1b). tiny, small and mid at three seeds, 1.2B tokens, ≈ 1.4 GPU-hours. Converts the surviving fixed-data ordering from uninterpretable to either claimed or retracted.

Candidate-set sweep variants (E4). The sweep itself is run: three wide-bank tasks, $|C| \in \{2, 8, 32, K\}$, at 113.47M (§4.10). What remains is a second seed at the same rung and the same twelve-arm grid at 33.7M and 14.4M — thirty-six arms at the measured 0.78 hours each, about 27.9 hours of wall time — plus staging those two checkpoints on the box. (18.6 hours for the seed and one rung.) The six narrow-bank tasks cannot pose $|C| = 32$ and are excluded structurally.

Context-length sweep (E5). Accuracy against context from 1k to 32k tokens, then to the design target. ≈ 1 GPU-hour at 32k.

Multimodal training. No result exists; this requires a data pipeline for quantised audio and a training run at matched tokens. It is a separate dissertation chapter and is scoped as one.

7.4 The claim, restated

A decider's structural guarantee is real and narrow: it cannot answer outside its candidate set. Whether it answers *correctly*, and whether that survives wider candidate sets and longer contexts, is the empirical question on which the commercial claim rests. This dissertation answers the width half at one rung: a trained 32-way interface is at or above the full bank on three wide-bank tasks, while a narrower interface is ahead of it by 5.80–13.85 points, so width is not free and the full bank is not a ceiling. The context half remains unmeasured, and one rung and one seed cannot establish that these three cells generalise. What the dissertation does establish is that the question was not previously measured under conditions that could support an answer, that it supplies the instrument which makes the measurement possible, and that it reports the width experiment it pre-registered, including where the result cuts against the stronger claim. The reclamation is part of the record: the numbers that survive are the ones with hashes, and the ones that do not are left visibly open.

Appendix A

Reproducing every number

Every table in this dissertation is emitted from result files by one of four generators: `thesis/gen_tables.py` for the training ladder, `thesis/gen_scaling.py` for the row-scaling and risk-coverage tables, `thesis/gen_comparator.py` for decode-versus-score and the CDN probe, and `thesis/gen_suite.py` for the public-task suite. Every inline figure in the prose is a macro from one of those generators. Nothing is transcribed by hand, because three numbers in this project's history were transcribed by hand and two of them were wrong: one could not be re-derived at all, and one split boundary was off by a factor of two because element counts were accumulated into a byte counter.

Table A.1: Provenance of every reported quantity.

Quantity	Command
Corpus build	`python3 scripts/build_corpus.py -target-bytes 1.6e9`
Ladder tables	`python3 thesis/gen_tables.py`
Row scaling and coverage	`python3 thesis/gen_scaling.py`
Decode versus score	`python3 thesis/gen_comparator.py`
Public-task suite	`python3 thesis/gen_suite.py`
Three-rung suite (Table 4.5)	`python3 thesis/gen_suite_three.py`
Exhaustive score	`python3 scripts/eval_full.py -run <dir> -rung <r>`
All rungs, one table	`python3 scripts/score_curve.py`
Segmented training	`python3 scripts/train_segments.py`
Milestone trajectory	`python3 scripts/track_curve.py`
Unattended plan	`python3 scripts/gpu_queue.py -plan research/queue_iso10.json`
Transfer matrix	`python3 scripts/transfer_sweep.py`
Sandbox re-stage	`scripts/molab_sync.sh -all`

A.1 Why the scorer is deliberately slow

The exhaustive scorer evaluates every held-out window in order rather than sampling, at a cost of 62–81 seconds per checkpoint. The sampled alternative produced the 0.3520

spread reported in §4.1, which is larger than the effect the ladder was designed to resolve. The exhaustive replicates that would quantify the reduction were lost with the first sandbox and are marked pending; the cost is stated here even though the comparison cannot be completed from this checkout.

A.2 Checkpoint identity and the reclamation

The first sandbox was reclaimed on 2026-10-28. The original AG News source checkpoint `3cf421...bed8` (28,973,877 bytes; step 6,104; 100,007,936 tokens) was lost with the disk and was never copied to the local checkout. The v2 corpus was rebuilt from the checked-in builder and reproduced its recorded SHA-256 exactly: `9267a2ea071a5178...`. The replacement checkpoint is `817e7f04...6140410` (28,973,877 bytes; step 6,104; 100,007,936 tokens; held-out 1.2937 bits/byte). It reproduces the protocol but not the bitwise weights, because GPU execution is not bitwise-deterministic.

The six AG News cells use the replacement checkpoint. The Banking77 84.3% row uses the surviving pre-loss comparator `499fd68c...24eebdc`. The two identities are not interchangeable, and no result in this thesis averages them into a single-checkpoint comparison.

The evidence tree `results/recovered_20261028/` is isolated from the canonical results tree and is accompanied by `MANIFEST.sha256`, which records the full digests (abbreviated above for the page) and hashes the corpus, the six AG News cells, the twenty-seven suite runs (nine tasks at three rungs) and the permutation controls. A second reclamation costs wall-clock, not results.

The public-task record spans two directories. `results/suite_dual/` holds the eighteen two-rung cells, their `SUITE.json` and their validation. `results/suite_large_113m/` holds the nine 113.47M cells, their pre-registration, the checkpoint record, the replication runs, the three-rung `COMPARISON.json` and a manifest of all 92 files. `thesis/gen_suite_three.py` reads the comparison, re-hashes every per-cell `result.json`, recomputes each accuracy from its own predictions, and asserts the 113.47M source-checkpoint digest before emitting `tables/suite3.tex`; a mismatch stops the build. The earlier `results/suite_final/` tree is retained only as *prior* metadata for the eight cells whose path changed and for the exact controls; it is not the current capability record.

A.3 Checkpoint discipline and the third rung

The lost 113.47M attempt is the reason this section exists. It completed 11 bounded segments, reached 980,746,240 of 1,134,658,560 tokens (86.44%) at step 59,860, and its sandbox was recycled mid-segment. `driver.json`, `run.json` and `train.jsonl` survived; no `ckpt.pt`, `checkpoint.json` or `segment.json` did. The training loop had saved weights at every boundary, but the puller licensed a weight pull only on a completion marker, so the last boundary weights were never copied; a fresh run had no step-0

checkpoint; and `-resume` did not verify the weights it loaded. The 980.7M-token offset is not a resumable model.

All four defects are fixed and covered by regression tests in `tests/test_pretrain_resume.py`: saves are atomic; a checkpoint is published at step 0; every checkpoint records the step, token offset, target, geometry, seed, RNG states and the weights SHA-256; `-resume` refuses to restart from zero, to change the budget or geometry, or to load an unloadable checkpoint; and `scripts/molab_pull.py` pulls on `checkpoint.json` alone and verifies the described byte count and digest. A second race surfaced on replay: the rolling `ckpt.pt` is replaced at the next boundary, so a slow puller can miss the boundary artifact. Each boundary is now archived to `checkpoints/seg_<index>_step_<step>/` and hash-verified before the next segment starts, and each artifact is pulled locally and verified before the following segment is launched. A sandbox death can therefore cost at most one segment.

The third rung is a *new* run under that discipline, not a continuation: fresh weights, step 69,255, 1,134,673,920 tokens, checkpoint aba65b60e30654a9…, staged from the local corpus whose digest matches the run record. Its evaluation ran in capped waves after the first nine-way wave exhausted the 97,887 MiB card: five cells fit and four died with CUDA OOM, and a later third co-residency exceeded the card. The failed waves produced no `result.json` and are preserved under `results/suite_large_113m/_failed_wave{1,2}/` with their verbatim reasons; they are infrastructure records, not measurements, and no cell was dropped or averaged across attempts.

A.4 Why a partial run cannot be graded

A run writes `summary.json` only after the held-out evaluation finishes. An interrupted rung therefore has no summary and cannot be averaged into a table as though it were complete. The driver additionally deletes the previous segment record before each run, because a completed segment was once reported as having reached 96,337,920 tokens when it had actually finished at 100,007,936: the record had been inherited from the segment before it. The reclamation record follows the same rule at the artifact level: a cell with no surviving summary is printed as pending, not reconstructed from the old PDF.

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